

General Theory of Renormalization of Gauge Invariant Operators

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We study the question of renormalization of gauge invariant operators in the gauge theories. Our discussion applies to gauge invariant operators of arbitrary dimensions and tensor structure. We show that the gauge noninvariant (and ghost) operators that mix with a given set of gauge invariant operators form a complete set of local solutions of a functional differential equation. We show that this set of gauge noninvariant operators together with the gauge invariant operators close under renormalization to all orders. We obtain a complete set of local solutions of the differential equation. The form of these solutions has recently been conjectured by Kluberg Stern and Zuber. With the help of our solutions, we show that there exists a basis of operators in which the gauge noninvariant operators “decouple” from the gauge invariant operators to all orders in the sense that *eigenvalues* corresponding to the eigenstates containing gauge invariant operators can be computed without having to compute the full renormalization matrix. We further discuss the substructure of the renormalization matrix.

I. INTRODUCTION

The problem of operator product expansion in gauge theories [1] has been studied extensively following the initial work of Georgi and Politzer [2], and Gross and Wilczek [3]. A salient feature of this problem, unique to gauge theories, is the possibility that the so-called Faddeev–Popov [4] ghost fields may participate in the operator product expansion of, say, two gauge invariant currents. Gross and Wilczek [3] dealt with this situation in the axial gauge, where the Faddeev–Popov ghost fields are absent (i.e., are free fields), and showed that the anomalous dimensions of gauge invariant operators are correctly given when possible couplings of the ghost fields are ignored in other gauges, at least in one-loop approximation. Subsequently, a number of authors, including Dixon and Taylor [5], Kluberg-Stern and Zuber [6, 7], Sarkar and Strubbe [8], have elaborated on and extended this result in some respects.

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The purpose of this paper is to give a general discussion of the renormalization of gauge invariant operators of arbitrary dimension and twist, and valid to any order of perturbation theory, along the lines exploited previously by Dixon and Taylor [5], and Kluberg-Stern and Zuber [6, 7]. In the course of this discussion, we will extend their results and prove conjectures made by some of the previous workers.

We base our discussion on the proof of renormalizability of gauge theories in the form presented by one of us [9] and streamlined by Zinn-Justin [10] by means of the Becchi–Rouet–Stora [11] (BRS) transformation. This is briefly summarized in Section II. The BRS transformation is a transformation of fields by an anti-commuting c -number λ , which leaves invariant the effective action $\mathcal{L}_{\text{eff}}$ defined in the gauge specified by f_α :

$$\begin{aligned}\delta A_i &= c_\alpha D_i^\alpha \lambda; & D_i^\alpha &= (\partial_i^\alpha + g t_{ij}^\alpha A_j), \\ \delta c_\alpha &= -\frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma \lambda, \\ \delta \bar{c}_\alpha &= -\eta_0^{1/2} f_\alpha \lambda,\end{aligned}\tag{1.1}$$

where A_i is the gauge field, c_α and $\bar{c}_\alpha$ are the Fadeev–Popov ghost fields, and g is the coupling constant.

In Section III, we show that only a subset of possible gauge noninvariant operators together with a set of gauge invariant operators of the same dimension and twist form a closed set under renormalization to all orders, and in Section IV, we give a complete characterization of the gauge noninvariant operators in this set. The problem reduces to finding a complete set of local functionals $H[\Phi, c, \partial\bar{c}, L]$ of a given dimension and ghost number that satisfy

$$\mathcal{G}' H = 0,\tag{1.2}$$

where the differential operator $\mathcal{G}'$ is

$$\mathcal{G}' = c_\alpha D_i^\alpha \frac{\delta}{\delta A_i} + \frac{1}{2} f_{\alpha\beta\gamma} c_\alpha c_\beta \frac{\delta}{\delta c_\gamma} - \frac{\delta \tilde{S}}{\delta A_i} \frac{\delta}{\delta (\partial\bar{c})_i} + \frac{\delta \tilde{S}}{\delta c_\alpha} \frac{\delta}{\delta L_\alpha},\tag{1.3}$$

L_α being the source of the operator $-\frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma$.

We solve Eq. (1.2) completely in Section IV, by a method suggested by Dixon and Taylor. Various mathematical lemmas necessary are proved in Appendices A and B. In all of these, the observation that

$$(\mathcal{G}')^2 = 0,\tag{1.4}$$

or more generally, that the BRS transformation on A_i and c_α is nilpotent, plays a crucial role. We confirm the conjecture of Kluberg–Stern and Zuber [7] on

solutions of Eq. (1.2), for arbitrary dimension and twist. Actually, the construction suggested by Dixon and Taylor does not make clear the locality of solution [12], but we have explicitly shown in Section IV and in Appendix C, the locality of solutions that is crucial to the arguments of Section V.

Section V is devoted to the study of the renormalization matrix of these operators. It is shown there that the renormalization matrix is in a block triangle form when the basis of operators closed under renormalization is appropriately chosen, and that in this basis, eigenvalues of the matrix corresponding to the eigenvectors containing gauge invariant operators are computable by neglecting couplings to gauge noninvariant operators.

We do not discuss the operator product expansion of gauge invariant currents per se in this paper, leaving it to a future communication.

II. REVIEW

A. Preliminary

In this section, we shall briefly review the definitions of generating functionals in a gauge theory, the BRS [11] (Becchi, Rouet, Stora) transformations and WT identities satisfied by the generating functionals. We shall use the condensed summation–integration convention as used, for example, in [9].

It is well known that the Feynman rules for constructing Green's functions of a gauge theory can be deduced from the effective action $\mathcal{L}_{\text{eff}}$.

$$\mathcal{L}_{\text{eff}}[A, c, \bar{c}] = \mathcal{L}_0[A] - \frac{1}{2}\{f_\alpha[A]\}^2 + \bar{c}_\alpha M_{\alpha\beta} c_\beta, \quad (2.1)$$

where $\mathcal{L}_0[A]$ is the Lagrangian for the Yang–Mills fields (possibly interacting with matter fields, in which case A denotes collectively gauge fields and matter fields), and

$$\mathcal{L}_g[A] \equiv -\frac{1}{2}\{f_\alpha[A]\}^2, \quad (2.2)$$

are the gauge fixing terms and $\bar{c}_\alpha, c_\beta$ are the Faddeev–Popov ghost fields.

The Lagrangian $\mathcal{L}_0[A]$ is invariant under the infinitesimal local gauge transformations of a compact Lie group G , which we shall choose to be a simple group solely for the sake of notational simplicity. Extension to the case of a direct product of simple compact groups is obvious.

Thus, $\mathcal{L}_0[A]$ is invariant under the infinitesimal transformation,

$$\begin{aligned} A_i \rightarrow A_i &= A_i + (\partial_i^\alpha + g_0 f_{ij}^\alpha A_j) \theta_\alpha \\ &\equiv A_i + D_i^\alpha[A] \theta_\alpha. \end{aligned} \quad (2.3)$$

The A_i 's are hermitian fields, so that the matrices $\{t^\alpha: (t^\alpha)_{ij} = t^\alpha_{ji}\}$ are real anti-symmetric representations of the generators of G . The g_0 is the (unrenormalized) coupling constant of the group G .

We shall work in linear gauges defined by,

$$f_\alpha[A] = \eta_0^{1/2} \partial_i^\alpha A_i, \quad (2.4)$$

where η_0 is an arbitrary real positive number.

Then, $M_{\alpha\beta}[A]$ of Eq. (2.1) is defined by

$$M_{\alpha\beta}[A] = \eta_0^{-1/2} \frac{\delta f_\alpha[A]}{\delta A_i} D_i^\beta[A]. \quad (2.5)$$

The BRS supertransformations consist of

$$\begin{aligned} \delta A_i &= c_\alpha D_i^\alpha \delta\lambda, \\ \delta c_\alpha &= -\frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma \delta\lambda, \\ \delta \bar{c}_\alpha &= -\eta_0^{1/2} f_\alpha[A] \delta\lambda, \end{aligned} \quad (2.6)$$

where $\delta\lambda$ is an x -independent infinitesimal anticommuting c -number. We note that under the BRS transformations

$$\begin{aligned} \delta(\mathcal{L}_0) &= 0; & \delta(c_\alpha D_i^\alpha) &= 0; & \delta(-\frac{1}{2} f_{\alpha\beta\gamma} c_\beta c_\gamma) &= 0; \\ \delta(-\frac{1}{2} f_\alpha^2[A] + \bar{c} M c) &= 0; & \delta(\mathcal{L}_{\text{eff}}[A, c, \bar{c}]) &= 0. \end{aligned} \quad (2.7)$$

We shall also find it useful to consider only the following transformations:

$$\begin{aligned} \delta A_i &= c_\alpha D_i^\alpha \delta\lambda; \\ \delta c_\alpha &= -\frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma \delta\lambda, & \delta \bar{c} &= 0. \end{aligned} \quad (2.8)$$

Under these transformations, $(\mathcal{L}_0[A] + \bar{c} M c)$ is invariant. This is expressed as

$$\mathcal{G}_0(\mathcal{L}_0[A] + \bar{c} M c) = 0, \quad (2.9)$$

where $\mathcal{G}_0$ is the differential operator defined by

$$\delta\{F[A, \bar{c}, c]\} = \mathcal{G}_0 F[A, c, \bar{c}] \delta\lambda, \quad (2.10a)$$

where δF refers to the change in the functional $F[A, c, \bar{c}]$ under the transformations of Eq. (2.8). If $F[A, c, \bar{c}]$ is a functional containing an equal number of c , $\bar{c}$ with all $\bar{c}$'s appearing before all c 's. $\mathcal{G}_0$ may be expressed as

$$\mathcal{G}_0 \equiv c_\alpha D_i^\alpha \frac{\delta}{\delta A_i} + \frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma \frac{\delta}{\delta c_\alpha}. \quad (2.10b)$$

In terms of $\mathcal{G}_0$, the group condition on $D_i^\alpha[A]$ viz.,

$$\left[D_i^\alpha \frac{\delta}{\delta \Phi_i}, D_i^\beta \frac{\delta}{\delta \Phi_i} \right] = g_0 f^{\alpha\beta\gamma} D_i^\gamma \frac{\delta}{\delta \Phi_i} \quad (2.11a)$$

i.e.,

$$t_{ij}^\alpha D_j^\beta - t_{ij}^\beta D_j^\alpha = f^{\alpha\beta\gamma} D_i^\gamma, \quad (2.11b)$$

can be expressed elegantly as the operator identity:

$$\mathcal{G}_0^2 \equiv 0. \quad (2.12)$$

B. Generating Functionals of a Gauge Theory; WT Identities

In the following, we shall be dealing with unrenormalized but dimensionally regularized quantities. Following Zinn-Justin, we introduce sources for the composite operators $c_\alpha D_i^\alpha$ and $\frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma$ and define:

$$S[A, c, \bar{c}, \kappa, l] = \mathcal{L}_{\text{eff}}[A, c, \bar{c}] + \kappa_i D_i^\alpha c_\alpha - \frac{1}{2} g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma l_\alpha. \quad (2.13)$$

We define the generating functional of Green's functions,

$$W[j, \xi, \bar{\xi}, \kappa, l] = \int [dA \, dc \, d\bar{c}] \text{Exp } i\{S[A, c, \bar{c}, \kappa, l] + j_i A_i + \bar{\xi}_\alpha c_\alpha + \bar{c}_\alpha \xi_\alpha\}, \quad (2.14)$$

where $\xi, \bar{\xi}$ are anticommuting sources for the ghost fields. The generating functional of the connected Green's functions is defined by,

$$Z[j, \xi, \bar{\xi}, \kappa, l] = -i \ln W[j, \xi, \bar{\xi}, \kappa, l]. \quad (2.15)$$

We define expectation values of fields in presence of sources:

$$\phi_i[j, \xi, \bar{\xi}, \kappa, l] = \frac{\delta Z}{\delta j_i}, \quad \Omega_\alpha = \frac{\delta Z}{\delta \xi_\alpha}; \quad \bar{\Omega}_\alpha = -\frac{\delta Z}{\delta \bar{\xi}_\alpha}, \quad (2.16)$$

where it is understood that the partial derivatives are taken with the rest of the sources fixed.

The generating functional of the proper vertices is given by

$$\Gamma[\phi, \Omega, \bar{\Omega}, \kappa, l] = Z[j, \xi, \bar{\xi}, \kappa, l] - j_i A_i - \bar{\xi}_\alpha c_\alpha - \bar{c}_\alpha \xi_\alpha. \quad (2.17)$$

The sources $j, \xi, \bar{\xi}$ can be expressed as a functional of $\phi, \Omega, \bar{\Omega}, \kappa, l$ by the relations:

$$j_i = -\frac{\delta \Gamma}{\delta \phi_i}, \quad \xi_\alpha = -\frac{\delta \Gamma}{\delta \Omega_\alpha}, \quad \bar{\xi}_\alpha = \frac{\delta \Gamma}{\delta \bar{\Omega}_\alpha}, \quad (2.18)$$

where as a matter of convention, partial derivatives of Γ such as $\delta\Gamma/\delta\phi_i$ stand for

$$\delta\Gamma/\delta\phi_i|_{\phi', \Omega, \bar{\Omega}, \kappa, l}. \quad (2.19)$$

With this convention, we also note,

$$\frac{\delta\Gamma}{\delta\kappa_i} = \frac{\delta Z}{\delta\kappa_i}, \quad \frac{\delta\Gamma}{\delta l_\alpha} = \frac{\delta Z}{\delta l_\alpha}. \quad (2.20)$$

By consideration of an arbitrary change $\delta\bar{c}_\alpha$ in the $W[j, \xi, \bar{\xi}, \kappa, l]$ of Eq. (2.14), we can obtain the equation of motion for the antighost field. It is expressed in terms of Γ as

$$\frac{\delta\Gamma}{\delta\bar{\Omega}_\alpha} = \partial_i^\alpha \frac{\delta\Gamma}{\delta\kappa_i}. \quad (2.21)$$

By considering the BRS transformations of Eq. (2.6) in the integration variables of W of Eq. (2.14), one can obtain the WT identity for W . It can be transformed into WT identity for Γ and simplified using Eq. (2.21). The final result is

$$\frac{\delta\Gamma_0}{\delta\phi_i} \frac{\delta\Gamma_0}{\delta\kappa_i} - \frac{\delta\Gamma_0}{\delta\bar{\Omega}_\alpha} \frac{\delta\Gamma_0}{\delta l_\alpha} = 0, \quad (2.22)$$

with

$$\Gamma_0[\phi, \Omega, \bar{\Omega}, \kappa, l] = \Gamma[\phi, \Omega, \bar{\Omega}, \kappa, l] + \frac{1}{2}\{f_\alpha[\phi]\}^2. \quad (2.23)$$

The theory is made finite by wave-function, and coupling constant renormalizations:

$$\begin{aligned} A_i &= A_i^{(r)} Z^{1/2}, & c_\alpha &= \tilde{Z}^{1/2} c_\alpha^{(r)}, & \bar{c}_\alpha &= \tilde{Z}^{1/2} \bar{c}_\alpha^{(r)}, \\ g_0 &= g X \tilde{Z}^{-1} Z^{-1/2}, \\ \kappa_i &= \tilde{Z}^{1/2} K_i, & l_\alpha &= Z^{1/2} L_\alpha. \end{aligned} \quad (2.24)$$

Alternatively, we can write the renormalized effective action as

$$\begin{aligned} S^{(r)}[A^{(r)}, c^{(r)}, \bar{c}^{(r)}, K, L] &= S[A, c, \bar{c}, \kappa, l] \\ &= S[A^{(r)}, c^{(r)}, \bar{c}^{(r)}, K, L] + \Delta S[A^{(r)}, c^{(r)}, \bar{c}^{(r)}, K, L], \end{aligned} \quad (2.25)$$

where ΔS represents local counterterms. It is a well-known result that Z , $\tilde{Z}$, and X in Eq. (2.24) can be chosen in successive loop approximations so that the resulting renormalized Γ is finite in terms of g and renormalized quantities.

III.

A. *Generating Functionals with an Insertion of a Local Operator*

In order to discuss renormalization of an operator, one needs to relate two quantities: Green's functions, involving an insertion of the operators that are independent of the renormalization point, the so-called unrenormalized Green's functions, and the finite (i.e., renormalized) Green's functions, which, however, depend on the renormalization point. In the following, we shall give the definitions for the generating functionals for these two Green's functions.

We consider a local (though it is not necessary for these definitions) operator $O_i[A, c, \bar{c}]$, which may carry additional Lorentz indices. Let N_i denotes (in general, x -dependent) source for $O_i[A, c, \bar{c}]$. The generating functional of the unrenormalized Green's functions with an arbitrary number of insertions of O_i is given by

$$W[j, \xi, \bar{\xi}, \kappa, l, N]$$

$$= \int [dA dc d\bar{c}] \exp i\{S[A, c, \bar{c}, \kappa, l] + j_i A_i + \bar{c}_\alpha \xi_\alpha + \bar{\xi}_\alpha c_\alpha + N_i O_i[A, c, \bar{c}]\}, \quad (3.1)$$

whose derivatives with respect to various sources are independent of the renormalization point for obvious reasons. The corresponding generating functionals of connected Green's functions and of proper vertices are defined analogs to Eqs. (2.15)–(2.18) of the last section viz.

$$Z[j, \xi, \bar{\xi}, \kappa, l, N] = -i \ln W[j, \xi, \bar{\xi}, \kappa, l, N], \quad (3.2a)$$

$$\Gamma[\phi, \Omega, \bar{\Omega}, \kappa, l, N] = Z[j, \xi, \bar{\xi}, \kappa, l, N] - j_i \phi_i - \bar{\Omega}_\alpha \xi_\alpha - \bar{\xi}_\alpha \Omega_\alpha. \quad (3.2b)$$

We further note that, as in Eq. (2.20)

$$\delta\Gamma/\delta N_i|_{\phi, \Omega, \bar{\Omega}, \kappa, l} = \delta Z/\delta N_i|_{j, \xi, \bar{\xi}, \kappa, l}. \quad (3.3)$$

The Fourier transform of the above quantity $\{\delta\Gamma/\delta N_i\}_{N=0=K=L}$ generates the unrenormalized proper vertices with a single insertion of O_i at an arbitrary momentum. Insertions at zero momentum can be obtained by considering $\int d^4x (\delta\Gamma/\delta N(x))_{N=K=L=0}$.

To renormalize

$$\Gamma_N[\phi, \Omega, \bar{\Omega}] = \delta\Gamma/\delta N_i|_{N=K=L=0} \quad (3.4)$$

in one loop approximation, one needs to compute $\{\Gamma_N\}_{\text{div}}^1$ expressed in terms of *renormalized* field Φ , ω , $\bar{\omega}$ and *renormalized* parameters: (renormalized according to some convention) and define

$$\{\Gamma_N^R[\Phi, \omega, \bar{\omega}]\}_1 = \{\Gamma_N[\phi, \Omega, \bar{\Omega}]\}_1 - \{\Gamma_N[Z^{1/2}\Phi, \bar{Z}^{1/2}\omega, \bar{Z}^{1/2}\bar{\omega}]\}_1^{\text{div}}, \quad (3.4)$$

$$\Phi = Z^{-1/2}\phi, \quad \omega = \bar{Z}^{-1/2}\Omega, \quad \bar{\omega} = \bar{Z}^{-1/2}\bar{\Omega}, \quad (3.4a)$$

so that Γ_N^R is finite in four space-time dimensions. Γ_N as a function of renormalized fields and parameters can be best obtained from another generating functional

$$W^{(r)}[J, \zeta, \bar{\zeta}, K, L, N] = \int [dA \, dc \, d\bar{c}] \exp i\{S^{(r)}[A, c, \bar{c}, K, L] + J_i A_i + \bar{\zeta}_\alpha c_\alpha + \bar{c}_\alpha \zeta_\alpha \\ + N_i O_i[Z^{1/2} A_i, \tilde{Z}^{1/2} c_\alpha, \tilde{Z}^{1/2} \bar{c}_\alpha, g(X/\tilde{Z} Z^{1/2}), \eta Z^{-1/2}]\}. \quad (3.5)$$

Here, the sources $J, \zeta, \bar{\zeta}, K, L$ stand for the renormalized sources. This $W^{(r)}$ is in fact equal to $W^{(u,r)}$ of Eq. (3.1). This can be seen by performing the transformations

$$j = Z^{-1/2} J, \quad \xi = \tilde{Z}^{-1/2} \zeta, \quad \bar{\xi} = \tilde{Z}^{-1/2} \bar{\zeta}, \quad \kappa = \tilde{Z}^{1/2} K, \quad l = Z^{1/2} L, \\ g_0 = gX/\tilde{Z} Z^{1/2}, \quad \eta_0 = \eta Z^{-1/2},$$

with a simultaneous change of integration variables

$$A_i = Z^{1/2} A^{(r)}_i, \quad c_\alpha = \tilde{Z}^{1/2} c^{(r)}_\alpha, \quad \bar{c}_\alpha = \tilde{Z}^{1/2} \bar{c}^{(r)}_\alpha,$$

and dropping the overall infinite constant and superscript (r) .

(We shall find it convenient to switch back and forth between $W^{(u,r)}$ and $W^{(r)}$. $W^{(u,r)}$ is especially useful to see the symmetry properties of W while it is $W^{(r)}$, which can give us $\{\Gamma_N\}^{\text{div}}$ as a functional of renormalized fields and parameters.)

B. WT Identity for the Generating Functional of Proper Vertices with a Single Insertion of a Gauge Invariant Operator

Our object here is to write down the WT identity satisfied by $\Gamma_N[\Phi, \omega, \bar{\omega}]$ in one loop approximation, when N refers to the source of a gauge invariant operator, and use it to find the properties of gauge noninvariant operators that enter in $\{\Gamma_N[\Phi, \omega, \bar{\omega}]\}_1^{\text{div}}$.

To do this, we consider the BRS transformations of Eq. (2.6) on the integration variables of $W[j, \xi, \bar{\xi}, \kappa, l, N]$ of Eq. (3.1), with $O_i[A, c, \bar{c}] \equiv O^{GI}[A]$. We note that the extra term $NO^{GI}[A]$ is invariant under these transformations. Therefore, the WT identity satisfied by $W[J, \zeta, \bar{\zeta}, K, L, N]$ is (i.e., expressed in terms of renormalized sources) and by $\Gamma[\Phi, \omega, \bar{\omega}, K, L, N]$ will be identical in form as the WT identities for these generating functionals at $N = 0$. Thus,

$$\frac{\delta \Gamma_0}{\delta \omega_\alpha} = \partial_i^\alpha \frac{\delta \Gamma_0}{\delta K_i}, \quad (3.6)$$

and

$$\frac{\delta \Gamma_0}{\delta \Phi_i} \frac{\delta \Gamma_0}{\delta K_i} - \frac{\delta \Gamma_0}{\delta \omega_\alpha} \frac{\delta \Gamma_0}{\delta L_\alpha} = 0. \quad (3.7)$$

We differentiate Eqs. (3.8) and (3.7) with respect to N and set $N = 0$. We thus obtain,

$$\frac{\delta \Gamma_N}{\delta \omega_\alpha} = \partial_i^\alpha \frac{\delta \Gamma_N}{\delta K_i} \quad (3.8)$$

$$\frac{\delta \Gamma_N}{\delta \Phi_i} \frac{\delta \Gamma_0}{\delta K_i} + \frac{\delta \Gamma_N}{\delta K_i} \frac{\delta \Gamma_0}{\delta \Phi_i} - \frac{\delta \Gamma_N}{\delta L_\alpha} \frac{\delta \Gamma_0}{\delta \omega_\alpha} - \frac{\delta \Gamma_N}{\delta \omega_\alpha} \frac{\delta \Gamma_0}{\delta L_\alpha} = 0, \quad (3.9)$$

with

$$\Gamma_N \equiv \Gamma_N[\Phi, \omega, \bar{\omega}, K, L] = \left. \frac{\delta \Gamma_0[\Phi, \omega, \bar{\omega}, K, L, N]}{\delta N} \right|_{N=0}. \quad (3.10)$$

Equating the one-loop divergence on both sides of the Eqs. (3.8) and (3.9), and remembering that $\Gamma_0|_{N=0}$ is a finite functional, we learn,

$$\frac{\delta}{\delta \omega_\alpha} \{\Gamma_N\}_1^{\text{div}} = \partial_i^\alpha \frac{\delta}{\delta K_i} \{\Gamma_N\}_1^{\text{div}} \quad (3.11)$$

$$\begin{aligned} & \left[\omega_\alpha D_i^\alpha \frac{\delta}{\delta \Phi_i} + \frac{1}{2} g f_{\alpha\beta\gamma} \omega_\beta \omega_\gamma \frac{\delta}{\delta \omega_\alpha} \right] \{\Gamma_N\}_1^{\text{div}} \\ &= - \frac{\delta \tilde{S}}{\delta \Phi_i} \frac{\delta}{\delta K_i} \{\Gamma_N\}_1^{\text{div}} + \frac{\delta \tilde{S}}{\delta \omega_\alpha} \frac{\delta}{\delta L_\alpha} \{\Gamma_N\}_1^{\text{div}}, \end{aligned} \quad (3.12)$$

with

$$\tilde{S}[\Phi, \omega, \bar{\omega}, K, L] = \mathcal{L}_0[\Phi] + \bar{\omega}_\alpha M_{\alpha\beta}[\Phi] \omega_\beta + K_i D_i^\alpha \omega_\alpha - \frac{1}{2} L_\alpha f_{\alpha\beta\gamma} \omega_\beta \omega_\gamma, \quad (3.13)$$

$\{\Gamma_N[\Phi, \omega, \bar{\omega}, K, L]\}_1^{\text{div}}$ is a local polynomial of $\Phi, \omega, \bar{\omega}, K, L$ of dimensions equal to that of $O^{\text{GI}}[\Phi]$. Thus, $\{\Gamma_N[\Phi, \omega, \bar{\omega}, K, L]\}_1^{\text{div}}$ can be expressed in terms of a complete set of local polynomial solutions of the equations:

$$\frac{\delta}{\delta \bar{\omega}_\alpha} O[\Phi, \omega, \bar{\omega}, K, L] = \partial_i^\alpha \frac{\delta}{\delta K_i} O[\Phi, \omega, \bar{\omega}, K, L], \quad (3.14)$$

$$\left[\mathcal{G}_0 - \frac{\delta \tilde{S}}{\delta \omega_\alpha} \frac{\delta}{\delta L_\alpha} + \frac{\delta \tilde{S}}{\delta \Phi_i} \frac{\delta}{\delta K_i} \right] O[\Phi, \omega, \bar{\omega}, K, L] = 0. \quad (3.15)$$

Thus, $\{\Gamma_N[\Phi, \omega, \bar{\omega}]\}_1^{\text{div}} = \{\Gamma_N[\Phi, \omega, \bar{\omega}, K, L]|_{K=L=0}\}_1^{\text{div}}$ can be expressed in terms of the complete set of local solutions of the equation

$$\frac{\delta O[\Phi, \omega, \bar{\omega}]}{\delta \bar{\omega}_\alpha} = \partial_i^\alpha Q_i[\Phi, \omega, \bar{\omega}], \quad (3.16)$$

and

$$\mathcal{G}_0 O[\Phi, \omega, \bar{\omega}] = -Q_i[\Phi, \omega, \bar{\omega}] \frac{\delta \tilde{S}}{\delta \Phi_i} + R_\alpha[\Phi, \omega, \bar{\omega}] \frac{\delta \tilde{S}}{\delta \omega_\alpha}, \quad (3.17)$$

where $Q_i[\Phi, \omega, \omega]$ and $R_\alpha[\Phi, \omega, \omega]$ are local functionals of Φ, ω, ω possessing appropriate transformation properties under the global transformations of group G and Lorentz transformations. (As we shall see later, the integrability of Eqs. (3.16) and (3.17) will restrict the set $\{Q_i\}$ and $\{R_\alpha\}$.)

We shall obtain the complete set of local solutions of the Eqs. (3.14) and (3.15) in the next section. In this section, we wish to show that the set of operators $\{O^G[\Phi]\} \oplus \{O[\Phi, \omega, \bar{\omega}]\}$ (the latter being the set of all gauge noninvariant solutions of Eqs. (3.16) and (3.17)); closes under renormalization to all orders. For this purpose, it is not necessary to know the set $\{O[\Phi, \omega, \bar{\omega}]\}$ explicitly.

C. Closure Property in One-Loop Approximation

We consider the generating functional of Green's functions W of Eq. (3.5) with $O_i[A, c, \bar{c}]$ being any one of the gauge noninvariant operators of the set. Our objective is to show that $\{\Gamma_N[\Phi, \omega, \bar{\omega}]\}_1^{\text{div}}$ thus computed for any of the gauge noninvariant operators can be expanded in terms of the same set $\{O^G[\Phi]\} \oplus \{O[\Phi, \omega, \bar{\omega}]\}$.

We shall find it convenient to use $W^{(u,r)}$ (in terms of unrenormalized sources) to exhibit its symmetry properties:

$$W^{(u,r)} = \int [dA \, dc \, d\bar{c}] \exp i\{\mathcal{L}_0[A] + \bar{c}Mc - \frac{1}{2}f_\rho^2[A] + \kappa_i' D_i^{\alpha'} c_\alpha - \frac{1}{2}f_\alpha' f_{\alpha\beta\gamma} c_\beta c_\gamma + N'O[A, c, \bar{c}] + j_i A_i + \bar{c}_\alpha \xi_\alpha + \xi_\alpha c_\alpha\}. \quad (3.18)$$

We note that $\{\mathcal{L}_0 + \bar{c}Mc - \frac{1}{2}f_\rho^2[A] + N'O[A, c, \bar{c}]\}$ is invariant under the transformations:

$$\begin{aligned} \delta A_i &= (D_i^\alpha c_\alpha + N'Q_i) \delta\lambda \equiv c_\alpha D_i^{\alpha'} \delta\lambda, \\ \delta c_\alpha &= (-\frac{1}{2}g_0 f_{\alpha\beta\gamma} c_\beta c_\gamma + N'R_\alpha) \delta\lambda \equiv -g_0 \frac{1}{2} f_{\alpha\beta\gamma}' c_\beta c_\gamma, \\ \delta \bar{c}_\alpha &= -\eta_0^{1/2} f_\alpha[A] \delta\lambda, \end{aligned} \quad (3.19)$$

to the first order in N' . Here, Q_i and R_α are the functions corresponding to $O[A, c, \bar{c}]$ that enter Eqs. (3.16) and (3.17). To see this, we note

$$\delta\{\mathcal{L}_{\text{eff}}[A, c, \bar{c}] + N'O[A, c, \bar{c}]\} = 0 + O(N') + O(N'^2) + \dots,$$

and $O(N')$ terms are given by,

$$N' \left\{ + \frac{\delta \mathcal{L}_{\text{eff}}}{\delta A_i} Q_i - \frac{\delta \mathcal{L}_{\text{eff}}}{\delta c_\alpha} R_\alpha \right\} \delta\lambda + N' \left\{ \mathcal{G}_0 O + \frac{\delta O}{\delta \bar{c}_\alpha} \eta_0^{1/2} f_\alpha \right\} \delta\lambda, \quad (3.20)$$

$$\begin{aligned} &= N' \left\{ + \frac{\delta \tilde{\mathcal{S}}}{\delta A_i} Q_i - \frac{\delta \tilde{\mathcal{S}}}{\delta c_\alpha} R_\alpha + \mathcal{G}_0 O \right\} \delta\lambda + N' \left\{ -\eta_0^{1/2} f_\alpha \partial_i^\alpha Q_i + \eta_0^{1/2} f_\alpha \frac{\delta O}{\delta \bar{c}_\alpha} \right\} \\ &= 0, \end{aligned} \quad (3.21)$$

on account of Eqs. (3.16) and (3.17). We further note that

$$\begin{aligned}\delta(D_i'^\alpha c_\alpha) &= O(N') \neq 0, \\ \delta(\tfrac{1}{2}f'_{\alpha\beta\gamma}c_\beta c_\gamma) &= O(N') \neq 0.\end{aligned}\tag{3.22}$$

We use the invariance of Eq. (3.21) under transformations of Eq. (3.19) to write down a WT identity of $W^{(u,r)}$ of Eq. (3.18). We note that the change in the Jacobian for the transformations of Eq. (3.19) is

$$\propto (Q_{i,i} - R_{\alpha,\alpha}) \propto \delta^4(x) \text{ or its derivatives at } x \equiv 0,$$

since Q_i and R_α are local functionals. We are using dimensional regularization, so that the Jacobian can be taken to be unity. We thus write the WT identity for $W^{(u,r)}$ (see Eq. (3.18)).

$$\begin{aligned}0 = \int [dA \, dc \, d\bar{c}] \{ &j_i D_i'^\alpha c_\alpha - \tfrac{1}{2} g_0 \bar{\xi}_\alpha f'_{\alpha\beta\gamma} c_\beta c_\gamma + \eta_0^{1/2} f'_\alpha [A] \xi_\alpha \\ &+ \text{terms of } O(\kappa' N', l' N', N'^2 \dots) \} \text{Exp } i\{\dots\}.\end{aligned}\tag{3.23}$$

We note that the WT identity of Eq. (3.23) is the same as the WT identity for the insertion of a gauge invariant operator, except for terms of $O(\kappa' N', l' N', N'^2, \dots)$. Further, by transformation of the antighost field $\bar{c}_\alpha \rightarrow \bar{c}_\alpha + \delta \bar{c}_\alpha$ (where $\delta \bar{c}_\alpha$ are independent of fields), we obtain the equation of motion for the antighost field:

$$0 = \int [dA \, dc \, d\bar{c}] \{ \partial_i^\alpha D_i'^\beta c_\beta + N' (\delta O / \delta \bar{c}_\alpha) + \xi_\alpha \} \exp i\{\dots\}.\tag{3.24}$$

Using Eq. (3.16) and the definition of $D_i'^\alpha [A]$, we may write Eq. (3.24) as

$$0 = \int [dA \, dc \, d\bar{c}] \{ \partial_i^\alpha D_i'^\beta c_\beta + \xi_\alpha \} \exp i\{\dots\}.\tag{3.25}$$

We can transcribe Eqs. (3.23) and (3.25) in terms of the renormalized sources:

$$0 = \left\{ \left[J_i \frac{\delta}{\delta K_i'} + \bar{\xi}_\alpha \frac{\delta}{\delta L_\alpha'} + \eta^{1/2} f'_\alpha \left[-i \frac{\delta}{\delta J} \right] \zeta_\alpha \right] W^{(r)}[J, \zeta, \bar{\xi}, K', L', N'] \right\}_{K'=L'=N'=0},\tag{3.26}$$

and

$$0 = \left[\partial_i^\alpha \frac{\delta}{\delta K_i'} + \zeta_\alpha \right] W^{(r)}[J, \zeta, \bar{\xi}, K', L', N'].\tag{3.27}$$

These can be translated in terms of the WT identity for proper vertices in the usual manner (and Eq. (3.26) simplified with the help of Eq. (3.27)) yielding,

$$\mathcal{G}_0 \{ \Gamma_{N'}[\Phi, \omega, \omega] \}_{11}^{\text{div}} = - \frac{\delta \tilde{\mathcal{S}}}{\delta \Phi_i} \left[\frac{\delta \Gamma_{N'}}{\delta K_i} \Big|_{K'=L'=0} \right]_1^{\text{div}} + \frac{\delta \tilde{\mathcal{S}}}{\delta \omega_\alpha} \left\{ \frac{\delta \Gamma_{N'}}{\delta L_\alpha} \Big|_{K=L=0} \right\}_1^{\text{div}},\tag{3.28}$$

and

$$\frac{\delta}{\delta \omega_\alpha} \{ \Gamma_{N'}[\Phi, \omega, \bar{\omega}]_1^{\text{div}} = \partial_i^\alpha \left\{ \frac{\delta \Gamma_{N'}}{\delta K_i} \right\}_{K=L=0}^{\text{div}}. \quad (3.29)$$

Therefore, it is clear that $\{ \Gamma_{N'}[\Phi, \omega, \bar{\omega}]_1^{\text{div}} \}$ is also expressible in terms of the complete set of solutions of Eqs. (3.16) and (3.17), i.e., $\{ O^G[\Phi] \} \oplus \{ O[\Phi, \omega, \bar{\omega}] \}$. This completes the proof of the closure property in one loop approximation.

D. Renormalization of Insertion of Operators and the Closure Property in Higher Orders

Next, we shall consider the renormalization of Green's functions (and proper vertices) with one insertion of operators in higher orders; and show that the closure property holds to all orders. To show the closure property, it is sufficient to know how to renormalize

$$\Gamma_N|_{K=0=L}, \quad \frac{\delta L_N}{\delta K} \Big|_{K=L=0}, \quad \frac{\delta \Gamma_N}{\delta L} \Big|_{K=L=0}.$$

We consider the complete set of independent triplets of functionals $\{ [O^{(p)}[\Phi, \omega, \bar{\omega}], Q_i^{(p)}[\Phi, \omega, \bar{\omega}], R_\alpha^{(p)}[\Phi, \omega, \bar{\omega}]] \}$ (and these include gauge invariant operators) such that each member (triplet) of the set satisfies Eqs. (3.16) and (3.17). (Some entries in the triplets may be zero, e.g., there may be more nonvanishing $Q_i^{(p)}$ than there are gauge noninvariant operators, etc.) Then,

$$\begin{aligned} \{ \Gamma_{N_p}[\Phi, \omega, \bar{\omega}]_1^{\text{div}} &= \sum_q z_{pq}^{(1)}(\epsilon) O^{(q)}[\Phi, \omega, \bar{\omega}], \\ \left\{ \frac{\delta \Gamma_{N_p}}{\delta K_i} \right\}_{K=L=0}^{\text{div}} &= \sum_q z_{pq}^{(1)}(\epsilon) Q_i^{(q)}[\Phi, \omega, \bar{\omega}], \\ \left\{ \frac{\delta \Gamma_{N_p}}{\delta L_\alpha} \right\}_{K=L=0}^{\text{div}} &= \sum_q z_{pq}^{(1)}(\epsilon) R_\alpha^{(q)}[\Phi, \omega, \bar{\omega}], \end{aligned} \quad (3.30)$$

(q runs over the complete set). Thus,

$$\begin{aligned} \{ \Gamma_{N_p}^{u,r}[\Phi, \omega, \bar{\omega}]_1 &= \{ Z_{pq} \Gamma_{N_p}^R[\Phi, \omega, \bar{\omega}]_1 \\ \left\{ \frac{\delta \Gamma_{N_p}^{u,r}}{\delta K_i} \right\}_{K=L=0} &= \left\{ Z_{pq} \frac{\delta \Gamma^R}{\delta K_i} \right\}_{K=L=0} \\ \left\{ \frac{\delta \Gamma_{N_p}^{u,r}}{\delta L_\alpha} \right\}_{K=L=0} &= \left\{ Z_{pq} \frac{\delta \Gamma^R}{\delta L_\alpha} \right\}_{K=L=0} \end{aligned} \quad (3.31)$$

where

$$Z_{pq} = \delta_{pq} + a \cdot z_{pq}^{(1)}, \quad (3.32)$$

where a is the loop expansion parameter.

The generating functional of Green's functions with a single operator insertion, in which all the internal subtractions up to one loop approximation (for the three types of Green's functions in Eq. (3.28)) have been performed, is expressed as

$$\begin{aligned} W[j, \xi, \bar{\xi}, \kappa', l', N_p] \\ = \int [dA \, dc \, d\bar{c}] \exp i\{\mathcal{L}_{\text{eff}} + j_i A_i + \bar{\xi}_\alpha \bar{c}_\alpha + c_\alpha \xi_\alpha + N_p Z_{pq} O^{(q)}[A, c, \bar{c}] \\ + \kappa'_i (D_i^\alpha c_\alpha + N_p Z_{pq} Q_i^{(q)} + l'_\alpha [-\tfrac{1}{2} f_{\alpha\beta\gamma} c_\beta c_\gamma + N_p Z_{pq} R_\alpha^{(q)}]\}. \end{aligned} \quad (3.33)$$

We have expressed W in terms of the unrenormalized sources only because the expression is simpler. The overall two-loop divergence is to be computed by expressing the corresponding generating functional for proper vertices in terms of renormalized quantities.

It is obvious that the definition of Eq. (3.31) can be extended to all orders simply by determining the renormalization matrix Z_{pq} in successive orders:

$$Z_{pq} = \delta_{pq} + a z_{pq}^{(1)} + a^2 z_{pq}^{(2)} + \dots, \quad (3.34)$$

where a is the loop expansion parameter, provided the counter terms needed are restricted to the set $\{[O^{(q)}, Q_i^{(q)}, R_\alpha^{(q)}]\}$; in other words, if the closure property holds to all orders. In the following, we shall show that this is true.

The proof proceeds by induction; and is very similar to the proof for the case of the one loop approximation. Let us assume that the closure property holds up to $(n-1)$ -loop approximation. Then, Z_{pq} is determined up to $(n-1)$ -loop approximation in Eq. (3.33). We note that under the transformation of integration variables,

$$\begin{aligned} \delta A_i &= (D_i^\alpha c_\alpha + N_p Z_{pq} Q_i^{(p)}) \delta \lambda, \\ \delta c_\alpha &= (-\tfrac{1}{2} g_{0\alpha\beta\gamma} c_\beta c_\gamma + N_p Z_{pq} R_\alpha^{(p)}) \delta \lambda, \\ \delta \bar{c}_\alpha &= -\eta_0^{1/2} f_\alpha[A] \delta \lambda, \end{aligned} \quad (3.35)$$

$$\delta\{\mathcal{L}_{\text{eff}}[A, c, \bar{c}] + N_p Z_{pq} O^{(p)}[A, c, \bar{c}]\} = O[N^2], \quad (3.36)$$

which is again a consequence of the fact that $\{O^{(p)}, Q_i^{(p)}, R_\alpha^{(p)}\}$ satisfy the Eqs. (3.14) and (3.15). One can derive the WT identity satisfied by $\Gamma[\Phi, \omega, \bar{\omega}, K, L, N]$. As

before, the WT identity satisfied by single insertions of any of the operators $\{O^{GI}\} \oplus \{O[A, c, \bar{c}]\}$ is the same viz.

$$\left\{ \frac{\delta \Gamma_N}{\delta \Phi_i} \frac{\delta \Gamma_0}{\delta K_{i'}} + \frac{\delta \Gamma_0}{\delta \Phi_i} \frac{\delta \Gamma_N}{\delta K_{i'}} - \frac{\delta \Gamma_0}{\delta \omega_\alpha} \frac{\delta \Gamma_N}{\delta L_\alpha} - \frac{\delta \Gamma_N}{\delta \omega_\alpha} \frac{\delta \Gamma_0}{\delta L_\alpha} \right\}_{K'=L'=0} = 0. \quad (3.37)$$

We equate the n -loop divergence on both sides noting

$$\left\{ \frac{\delta^2 \Gamma_N}{\delta K_{i'}} \right\}_{r\text{-loop}}^{K'=L'=N=0} = \text{finite} \quad \left\{ \frac{\delta \Gamma_N}{\delta L_\alpha} \right\}_{r\text{-loop}}^{K'=L'=N=0} = \text{finite}, \quad r \leq n-1, \quad (3.38)$$

we get

$$0 = \left[\left\{ \frac{\delta \Gamma_0}{\delta K_{i'}} \right\}_0 \left\{ \frac{\delta \Gamma_N}{\delta \Phi_i} \right\}_n^{\text{div}} + \left\{ \frac{\delta \Gamma_0}{\delta \Phi_i} \right\}_0 \left\{ \frac{\delta \Gamma_N}{\delta K_{i'}} \right\}_n^{\text{div}} - \left\{ \frac{\delta \Gamma_0}{\delta \omega_\alpha} \right\}_0 \left\{ \frac{\delta \Gamma_N}{\delta L_\alpha} \right\}_n^{\text{div}} - \left\{ \frac{\delta \Gamma_0}{\delta L_\alpha} \right\}_0 \left\{ \frac{\delta \Gamma_N}{\delta \omega_\alpha} \right\}_n^{\text{div}} \right]_{K'=L'=0}, \quad (3.39)$$

and thus

$$\mathcal{G}_0 \{ \Gamma_N \}_n^{\text{div}} = - \frac{\delta \tilde{S}}{\delta \Phi_i} \left\{ \frac{\delta \Gamma_N}{\delta K_{i'}} \right\}_{K'=L'=0}^{\text{div}}_n - \frac{\delta \tilde{S}}{\delta \omega_\alpha} \left\{ \frac{\delta \Gamma_N}{\delta L_\alpha} \right\}_{K'=L'=0}^{\text{div}}_n. \quad (3.40)$$

Similarly, the equation of motion for the $\bar{c}$ field has the same form

$$\frac{\delta}{\delta \omega_\alpha} [\Gamma_N]_n^{\text{div}} = \partial_i^\alpha \left\{ \frac{\delta \Gamma_N}{\delta K_{i'}} \right\}_{K'=L'=0}^{\text{div}}_n. \quad (3.41)$$

Thus, it is clear that $\{ \Gamma_N[\Phi, \omega, \bar{\omega}] \}_n^{\text{div}}$ can be expressed in terms of the same set of operators $\{O^{GI}[\Phi]\} \oplus \{O[\Phi, \omega, \bar{\omega}]\}$. Thus, the proof by induction is complete.

IV. FORM OF THE GAUGE NONINVARIANT OPERATORS

In this section, we shall show that the complete set of gauge noninvariant local solutions [14] of the Eqs. (3.16) and (3.17) can be written in the form

$$O[\Phi, \omega, \bar{\omega}] = \frac{\delta S[\Phi, \omega, \bar{\omega}]}{\delta \Phi_i} \frac{\delta F[\Phi, \omega, \bar{\omega}]}{\delta (\partial \bar{\omega})_i} + \frac{\delta S[\Phi, \omega, \bar{\omega}]}{\delta \omega_\alpha} X_\alpha[\Phi, \omega, \bar{\omega}] + \mathcal{G}_0[\Phi, \omega, \bar{\omega}] F[\Phi, \omega, \bar{\omega}], \quad (4.1)$$

where $F[\Phi, \omega, \bar{\omega}]$ and $X_\alpha[\Phi, \omega, \bar{\omega}]$ are arbitrary local polynomials of $\Phi, \omega, \bar{\omega}$ possessing the appropriate ghost number and appropriate transformation proper-

ties under global transformations of G and under Lorentz transformations. This form of the operators $\{O[\Phi, \omega, \bar{\omega}]\}$ will be the basis of our proof of "decoupling" in the next section.

Terms in $Q_i[\Phi, \omega, \bar{\omega}, K, L]$ that vanish when multiplied by ∂_i^α do not have a counter part in $O[\Phi, \omega, \bar{\omega}, K, L]$ as seen from Eq. (3.14). Therefore, we may choose to write $O[\Phi, \omega, \bar{\omega}, K, L]$ such that

$$\frac{\delta O[\Phi, \omega, \bar{\omega}, K, L]}{\delta(\partial\bar{\omega})_i} = Q_i[\Phi, \omega, \bar{\omega}, K, L]. \quad (4.2)$$

We shall find it convenient to consider Eqs. (3.14) and (3.15) at $K = 0$ instead of Eqs. (3.16) and (3.17). Replacing Q_i in Eq. (3.15) by the expression (4.2) and setting $K = 0$, we obtain

$$\left\{ \mathcal{G}_0 - \frac{\delta\tilde{S}}{\delta\omega_\alpha} \frac{\delta}{\delta L_\alpha} + \frac{\delta\tilde{S}}{\delta\Phi_i} \frac{\delta}{\delta(\partial\bar{\omega})_i} \right\} O[\Phi, \omega, \bar{\omega}, K, L] = 0. \quad (4.3)$$

We define a new differential operator

$$\mathcal{G}' = \mathcal{G}_0 - \frac{\delta\tilde{S}}{\delta\omega_\alpha} \frac{\delta}{\delta L_\alpha} + \frac{\delta\tilde{S}}{\delta\Phi_i} \frac{\delta}{\delta(\partial\bar{\omega})_i}, \quad (4.4)$$

which can be easily shown to satisfy

$$\mathcal{G}'^2 = 0. \quad (4.5)$$

It is easy to see that all functionals of the form $\mathcal{G}'F[\Phi, \omega, \bar{\omega}, L]$ are solutions of Eq. (4.3). It is also easy to see that all gauge invariant operators are solutions of Eq. (4.3). Our objective is to show that the two types of solutions are the *only local* solutions of Eq. (4.3). Thus, we state our theorem:

MAIN THEOREM. *The complete set of independent local solutions of the Eq. (4.3) with ghost number zero can be expressed as*

$$\{\mathcal{G}'F[\Phi, \omega, \bar{\omega}]\} \oplus \{O^{GI}[\Phi]\}',$$

where $F[\Phi, \omega, \bar{\omega}]$ are arbitrary local polynomials with ghost number (-1) . Here, the ghost number is defined as the difference between the powers of ω and $\bar{\omega}$ in the expression. $\{O^{GI}[\Phi]\}'$ refers to the set of gauge invariant operators not expressible as $\mathcal{G}'F$, i.e., as $(\delta\mathcal{L}/\delta\Phi_i)F_i'$ [14].

We shall prove first a lemma which we shall find very useful in the proof. The method is essentially similar to that employed by Dixon and Taylor.

LEMMA 1 (Dixon–Taylor). *Let $\mathcal{G}$ be a local differential operator in $\Phi, \omega, \bar{\omega}, L$. Let it be possible to expand $\mathcal{G}$ in powers of some parameter β as*

$$\mathcal{G}(\beta) = A + \beta B + \beta^2 C, \quad (4.6)$$

and let $\mathcal{G}(\beta)$ satisfy,

$$\mathcal{G}^2(\beta) = 0, \quad \text{for arbitrary } \beta. \quad (4.7)$$

Then, all polynomial solutions of the equation

$$\mathcal{G}(\beta) H(\beta) = 0, \quad (4.8)$$

where $H(\beta)$ can be expanded as a power series in β :

$$H(\beta) = \sum_{n=0} H^{(n)} \beta^n, \quad (4.9)$$

can be expressed as

$$H(\beta) = \mathcal{G}(\beta) J(\beta), \quad J(\beta) = \sum_{n=0} \beta^n J^{(n)}, \quad (4.10)$$

provided that the equation $AY = 0$ implies that there exists a X such that

$$Y = AX, \quad (4.11)$$

where Y has the same quantum numbers as H .

Proof. Equation (4.7) can be written as

$$A^2 = 0, \quad (4.12a)$$

$$AB + BA = 0, \quad (4.12b)$$

$$CA + AC + B^2 = 0, \quad (4.12c)$$

$$BC + CB = 0, \quad (4.12d)$$

$$C^2 = 0. \quad (4.12e)$$

Equation (4.9) gives,

$$AH^{(n)} + BH^{(n-1)} + CH^{(n-2)} = 0. \quad (4.13)$$

We have to show that $J^{(n)}$ exists such that

$$H^{(n)} = AJ^{(n)} + BJ^{(n-1)} + CJ^{(n-2)}. \quad (4.14)$$

Let us assume that we have determined $J^{(n)}$ for $n < r$. We thus know that

$$\begin{aligned} H^{(r-1)} &= AJ^{(r-1)} + BJ^{(r-2)} + CJ^{(r-3)}, \\ H^{(r-2)} &= AJ^{(r-2)} + BJ^{(r-3)} + CJ^{(r-4)}. \end{aligned} \quad (4.15)$$

Then, Eq. (4.13) yields,

$$AH^{(r)} = -B[AJ^{(r-1)} + BJ^{(r-2)} + CJ^{(r-3)}] - C[AJ^{(r-2)} + BJ^{(r-3)} + CJ^{(r-4)}]. \quad (4.16)$$

Using Eqs. (4.12) in Eq. (4.16),

$$AH^{(r)} = ABJ^{(r-1)} + ACJ^{(r-2)},$$

i.e.,

$$A[H^{(r)} - BJ^{(r-1)} - CJ^{(r-2)}] = 0. \quad (4.17)$$

We then can determine the desired $J^{(r)}$ if the Eq. (4.17) implies that there exists $J^{(r)}$ such that

$$H^{(r)} - BJ^{(r-1)} - CJ^{(r-2)} = AJ^{(r)}. \quad (4.18)$$

Applying the same argument for $r = 0$, we learn that we can determine $J^{(0)}$ if the equation

$$AH^{(0)} = 0 \quad (4.19)$$

implies that there exists a $J^{(0)}$ such that

$$H^{(0)} = AJ^{(0)}. \quad (4.20)$$

Thus, Eqs. (4.19) and (4.20), and Eqs. (4.17) and (4.18) imply that $J^{(0)}$ and $J^{(r)}$ $r \geq 1$ can be determined successively if the equation $AY = 0$ implies that there exists an X such that $Y = AX$, proving the lemma.

Comments. (i) If Eq. (4.11) holds, $J(\beta)$ can be found. However, this $J(\beta)$ is not unique, for we may add to $J(\beta)$ a polynomial $K(\beta)$ such that

$$\mathcal{G}(\beta) K(\beta) = 0. \quad (4.21)$$

In particular, $K(\beta)$ need not start as β^0 , it may start as some positive or some negative integral power of β . If $J(\beta)$ exists,

$$J'(\beta) = J(\beta) + K(\beta)$$

is the most general expression satisfying $\mathcal{G}J' = H$. Therefore, there is no loss of generality in assuming that $J(\beta)$ starts as β^0 .

(ii) We may use special cases of the lemma such as putting $C = 0$.

(iii) Given that H is local, the lemma does not make any statement as to whether J can be chosen to be local.

Now, let us return to the Eq. (4.3), which we have to solve. Let us expand a local solution $O^{(p)}[\Phi, \omega, \bar{\omega}]$

$$O^{(p)}[\Phi, \omega, \bar{\omega}, L] = F^{(p)}[\Phi] + (\partial\bar{\omega})_i Q_i^{(p)}[\Phi, \omega, \bar{\omega}] + \text{terms proportional to } L. \quad (4.22)$$

(We have used Eq. (3.14).) We note that the terms proportional to L involve at least two factors of ω . Substituting Eq. (4.22) in Eq. (4.3) and comparing the coefficients of the lowest power in ω , $\bar{\omega}$ (i.e., $(\bar{\omega})^0 \omega^1$), we get

$$D_i^\alpha \frac{\delta F[\Phi]}{\delta \Phi_i} = Q_i^\alpha[\Phi] \mathcal{L}_{0,i}[\Phi], \quad (4.23)$$

where,

$$Q_i^\alpha[\Phi] = \left. \frac{\delta Q_i}{\delta \omega_\alpha} [\Phi, \omega, \bar{\omega}] \right|_{\omega=\bar{\omega}=0}. \quad (4.24)$$

Our method will be

- (i) To show that all particular solutions of Eq. (4.23) can be expressed as

$$F[\Phi] = \mathcal{L}_{0,i} S_i[\Phi], \quad (4.25)$$

$S_i[\Phi]$ being an arbitrary local functional (Theorem I).

- (ii) To isolate all the solutions of Eq. (4.3) that must have a nonzero $F[\Phi]$ in Eq. (4.22). The rest of the independent solutions may be assumed to contain at least one ghost (ω).

- (iii) To solve for such solutions (containing at least one ghost) by using the Dixon–Taylor lemma in two stages:

Step I. We write Eq. (4.23) as

$$\omega_\alpha D_i^\alpha \frac{\delta F[\Phi]}{\delta \Phi_i} = \mathcal{G}_0 F[\Phi] = \omega_\alpha Q_i^\alpha \mathcal{L}_{0,i}, \quad (4.26)$$

where $\mathcal{G}_0$ is defined in Eq. (2.10) and satisfies $\mathcal{G}_0^2 = 0$. We thus have

$$\mathcal{G}_0(\omega_\alpha Q_i^\alpha \mathcal{L}_{0,i}) = 0. \quad (4.27)$$

We define an operator

$$\mathcal{G} = \mathcal{G}_0 - \text{covariance terms}, \quad (4.28)$$

where covariant terms are obtained by “varying” the free indices of the functional on which it acts according to the following law:

$$\begin{aligned} (\text{covariance terms}) A_i[\Phi, \omega, \bar{\omega}] &= -g t_{ij}^\alpha \omega_\alpha A_j[\Phi, \omega, \bar{\omega}], \\ (\text{covariance terms}) A_\alpha[\Phi, \omega, \omega] &= -g f_{\alpha\beta\gamma} \omega_\beta A_\gamma[\Phi, \omega, \bar{\omega}]. \end{aligned} \quad (4.29)$$

It can be shown that, in general,

$$\mathcal{G}^2 = 0. \quad (4.30)$$

We note that

$$\mathcal{G}[\mathcal{L}_{0,i}] \equiv \mathcal{G}_0 \mathcal{L}_{0,i} - g t_{ij}^\alpha \omega_\alpha \mathcal{L}_{0,j} = 0. \quad (4.31)$$

Using Eq. (4.31), Eq. (4.27) can be written as

$$\mathcal{G}(\omega_\alpha Q_i^\alpha[\Phi]) \mathcal{L}_{0,i} = 0. \quad (4.32)$$

This is the integrability condition for a solution of Eq. (4.23) to exist given Q_i^α .

Using Lemma A2 in Appendix A, we infer from Eq. (4.32) that

$$\mathcal{G}(\omega_\alpha Q_i^\alpha[\Phi]) = \omega_\alpha \omega_\beta X_{\alpha\beta\gamma}[\Phi] D_i^\gamma + \omega_\alpha \omega_\beta Y_{ij}^{\alpha\beta}[\Phi] \mathcal{L}_{0,j}, \quad (4.33)$$

where

$$Y_{i_1 i_2}^{\alpha\beta}[\Phi] = \sum_r Y_{i_1 i_2 \dots i_{r+2}}^{\alpha\beta(\gamma)}[\Phi] \mathcal{L}_{0,i_3} \dots \mathcal{L}_{0,i_{r+2}}, \quad (4.34)$$

with

$$\sum_{\{n\}} Y_{i_{n_1} \dots i_{n_{r+2}}}^{\alpha\beta(r)} = 0. \quad (4.35)$$

It is shown in the Appendix B that Eq. (4.33) implies that there exist local functions T and U such that

$$\omega_\alpha Q_i^\alpha = \omega_\alpha T_{\alpha\beta} D_i^\beta + \omega_\alpha U_{ij}^\alpha \mathcal{L}_{0,j} + \text{solutions of } \mathcal{G}(\omega_\alpha Q_i^\alpha[\Phi]) = 0, \quad (4.36)$$

where

$$U_{i_1 i_2}^\alpha[\Phi] = \sum_{r=0} U_{i_1 i_2 \dots i_{r+2}}^{\alpha(r)} \mathcal{L}_{0,i_3} \dots \mathcal{L}_{0,i_{r+2}} \quad (4.37)$$

$$\sum_{\{n\}} U_{i_{n_1} \dots i_{n_{r+2}}}^{\alpha(r)} = 0. \quad (4.38)$$

Therefore, the set of $Q_i^\alpha[\Phi]$'s for which a solution to Eq. (4.23) exists, and hence, satisfies Eq. (4.32), can be divided into two classes

$$(i) \quad \{Q_i^\alpha\}_{\text{class 1}} = \{T_{\alpha\beta} D_i^\beta\} \oplus \{U_{ij}^\alpha \mathcal{L}_{0,j}\}. \quad (4.39)$$

They satisfy

$$Q_i^\alpha(\text{class 1}) \mathcal{L}_{0,i} = 0, \quad (4.40)$$

and therefore, a particular solution $F[\Phi]$ for Q_i^α in class 1 can be taken to be zero.

(ii) $Q_i^\alpha[\Phi]$, which satisfy

$$\mathcal{G}(\omega_\alpha Q_i^\alpha) = 0. \quad (4.41)$$

As shown by Dixon and Taylor, for each such $Q_i^\alpha[\Phi]$ satisfying Eq. (4.41), there exists a polynomial $S_i[\Phi]$ such that

$$\omega_\alpha Q_i^\alpha = \mathcal{G} S_i[\Phi]. \quad (4.42)$$

(These follow essentially by the use of the Dixon Taylor lemma for the operator $\mathcal{G}$ of Eq. (4.28).) It is shown in Appendix C that $S_i[\Phi]$ can be chosen to be local. Therefore, a particular solution of Eq. (4.23) may be chosen to be

$$F[\Phi] = \mathcal{L}_{0,i} S_i[\Phi]. \quad (4.43)$$

We also note that the particular solution of Eq. (4.23) is unique modulo gauge invariant functionals.

Thus, we have proved Theorem I.

Step II. We note that because of the property $\mathcal{G}'^2 = 0$, $\mathcal{G}' S_i^{(p)}[\Phi](\partial\bar{\omega})_i$ is a solution of Eq. (4.3). Therefore, we separate independent solutions of Eq. (4.3) in to two classes:

$$\{O^{(p)}[\Phi, \omega, \bar{\omega}, L]\} = \{\mathcal{G}' S_i^{(p)}[\Phi](\partial\bar{\omega})_i\} \oplus \{O^{(p)}[\Phi, \omega, \bar{\omega}, L] - \mathcal{G}' S_i^{(p)}[\Phi](\partial\bar{\omega})_i\}, \quad (4.44)$$

where $S_i^{(p)}$ is related to $F^{(p)}$ of Eq. (4.22) by Eq. (4.43).

We note that

$$\begin{aligned} \mathcal{G}' S_i^{(p)}[\Phi](\partial\bar{\omega})_i &= \mathcal{L}_{0,i}[\Phi] S_i^{(p)}[\Phi] + g(\partial\bar{\omega})_j t_{ji}^\alpha \omega_\alpha S_i^{(p)} + \mathcal{G}_0(\partial\bar{\omega})_i S_i^{(p)} \\ &= F^{(p)}[\Phi] - (\partial\bar{\omega})_i [\mathcal{G} S_i^{(p)}]. \end{aligned}$$

Thus, we see that

(i) $O^{(p)}[\Phi, \omega, \bar{\omega}, L] - \mathcal{G}' S_i^{(p)}[\Phi]$ does not contain terms independent of $\omega, \bar{\omega}$.

(ii) The terms linear in $\bar{\omega}, \omega$ are

$$(\partial\bar{\omega})_i [Q_i^{\alpha(p)}[\Phi] \omega_\alpha - \mathcal{G} S_i^{(p)}[\Phi]],$$

which are expressible as $(\partial\bar{\omega})_i (T_{\alpha\beta}^{(p)} D_i^\beta + U_{ij}^{\alpha(p)} \mathcal{L}_{0,j}) \omega_\alpha$ by virtue of Eqs. (4.39) and (4.42).

The problem is thus, reduced to finding solutions of Eq. (4.3) that contain at least one factor of ω_α .

Step III. We note that the operator $\mathcal{G}'$ can be expressed as

$$\mathcal{G}' = A + gB + g^2C, \quad (4.45)$$

where

$$A = \omega_n \partial_k^\eta \frac{\delta}{\delta \Phi_k} - D_{ij} \Phi_j \frac{\delta}{\delta (\partial \bar{\omega})_i} + (\partial \bar{\omega})_i \partial_i^\alpha \frac{\delta}{\delta L_\alpha}, \quad (4.46)$$

$$D_{ij} \Phi_j = \mathcal{L}_{0,i} \big|_{g=0}, \quad (4.47)$$

and $\mathcal{G}'$ satisfies $\mathcal{G}'^2 = 0$.

Solutions of our interest can be expanded in powers of g (we may assume that the expansion begins as g^0)

$$O[\Phi, \omega, \bar{\omega}, L] = \sum g^n O^{(n)}[\Phi, \omega, \bar{\omega}, L]. \quad (4.48)$$

The Dixon–Taylor lemma tells us that there exists a functional $F[\Phi, \omega, \bar{\omega}, L]$ such that,

$$O[\Phi, \omega, \bar{\omega}, L] = \mathcal{G}' F[\Phi, \omega, \bar{\omega}, L], \quad (4.49)$$

if the equation ($Y[\Phi, \omega, \bar{\omega}, L]$ has the same quantum numbers as O)

$$AY[\Phi, \omega, \bar{\omega}, L] = 0, \quad (4.50)$$

implies that there exists a functional $X[\Phi, \omega, \bar{\omega}, L]$ such that

$$Y[\Phi, \omega, \bar{\omega}, L] = AX[\Phi, \omega, \bar{\omega}, L]. \quad (4.51)$$

Thus, we have to prove Eq. (4.51) given Eq. (4.50). To do this, we consider a scaling transformation on ω and L :

$$\bar{\omega} = \alpha \bar{\omega}', \quad L = \alpha L', \quad \Phi' = \Phi, \quad \omega' = \omega. \quad (4.52)$$

Then, in terms of the new variables Eq. (4.50) becomes (dropping primes)

$$(A_0 + (1/\alpha) A_1) Y[\Phi, \omega, \alpha \bar{\omega}, \alpha^2 L] = 0, \quad (4.53)$$

where

$$A_0 = \omega_n \partial_k^\eta (\delta / \delta \Phi_k), \quad (4.54)$$

$$A_1 = -D_{ij} \Phi_j \frac{\delta}{\delta (\partial \bar{\omega})_i} + (\partial \bar{\omega})_i \partial_i^\alpha \frac{\delta}{\delta L_\alpha}. \quad (4.55)$$

We note that in Eq. (4.50), we need to consider $Y[\Phi, \omega, \bar{\omega}, L]$, which are local polynomials in $\bar{\omega}$ and L , and therefore, $Y[\Phi, \omega, \alpha \bar{\omega}, \alpha^2 L]$ contains a finite maximum power of α , which may be determined given the dimension of the operators we are interested in. Multiplying Eq. (4.53) by α , we get

$$(A_1 + \alpha A_0) \{Y[\Phi, \omega, \alpha \bar{\omega}, \alpha^2 L]\} = 0. \quad (4.56)$$

We further note that,

$$(A_1 + \alpha A_0)^2 = 0, \quad \text{for any } \alpha. \quad (4.57)$$

Thus, we may apply the Dixon–Taylor lemma to Eq. (4.56) and deduce that there exists a *local* polynomial $Y[\Phi, \omega, \bar{\omega}, L, \alpha]$ such that

$$Y[\Phi, \omega, \alpha \bar{\omega}, \alpha^2 L] = (A_1 + \alpha A_0) X[\Phi, \omega, \bar{\omega}, L, \alpha], \quad (4.58)$$

provided for any *local* functional $Z[\Phi, \omega, \bar{\omega}, L]$ with the same quantum numbers as Y {and therefore, $O[\Phi, \omega, \bar{\omega}, L]$ } satisfying

$$A_1 Z[\Phi, \omega, \partial \bar{\omega}, L] = 0, \quad (4.59)$$

implies that there exists a *local* functional $V[\Phi, \omega, \partial \bar{\omega}, L]$ such that

$$Z[\Phi, \omega, \partial \bar{\omega}, L] = A_1 V[\Phi, \omega, \partial \bar{\omega}, L]. \quad (4.60)$$

Since $O[\Phi, \omega, \partial \bar{\omega}, L]$ is proportional to at least one ω , it must be Z of Eq. (4.59). (At this point we note that we could have written Eq. (4.53) as

$$\gamma^N (A_0 + \gamma A_1) Y[\Phi, \omega, \bar{\omega}/\gamma, L/\gamma^2] = 0 \quad (4.61)$$

with $\gamma = 1/\alpha$ and N is some finite integer; and attempted proving statement of Eq. (4.60) for A_0 instead of A_1 . However since,

$$\begin{aligned} A_0 &= \omega_\eta \partial_i^n \frac{\delta}{\delta \Phi_\eta} \\ &= \omega_\eta \frac{\delta}{\delta \Phi_\eta^L}, \end{aligned} \quad (4.62)$$

where Φ_η^L is the longitudinal gauge field defined by,

$$\Phi_i = \Phi_i^T + \partial_i^n \Phi_\eta^L; \quad \partial_i^\alpha \Phi_i^T = 0, \quad (4.63)$$

and thus, involves a derivative with respect to Φ^L that is not a local functional of Φ . Thus, even though a statement analogous to Eq. (4.60) may hold, it is difficult to decide whether the corresponding V can be chosen to be local. In fact, there are instances when a V exists but can never be chosen to be local.)

To prove Eq. (4.60) given Eq. (4.59), we expand Z in powers of L :

$$\begin{aligned} Z[\Phi, \omega, \partial \bar{\omega}, L] &= \sum_{n=0}^m Z^{(n)}, \\ Z^{(n)} &= \frac{1}{n!} Z^{(n)\alpha_1 \dots \alpha_n} L_{\alpha_1} \dots L_{\alpha_n}. \end{aligned} \quad (4.64)$$

Compare the coefficients $(L)^n$ on both sides of Eq. (4.59):

$$(D\Phi)_i \frac{\delta}{\delta(\partial\bar{\omega})_i} Z^{(n)} + (\partial\bar{\omega})_i \partial_i^\alpha \frac{\delta Z^{(n+1)}}{\delta L_\alpha} = 0, \quad 0 \leq n \leq m-1, \quad (4.65)$$

and

$$(D\Phi)_{i\delta(\partial\bar{\omega})_i} Z^{(m)\alpha_1 \cdots \alpha_m} = 0. \quad (4.66)$$

In Eq. (4.64), we can always assume $Z^{(m)\alpha_1 \cdots \alpha_n}$ to be completely symmetric in $\alpha_1 \cdots \alpha_n$:

$$S[\alpha_1 \cdots \alpha_n] Z^{(n)\alpha_1 \cdots \alpha_n} = Z^{(n)\alpha_1 \cdots \alpha_n}. \quad (4.67)$$

We will begin with Eq. (4.66). We expand $Z^{(m)}$ further in $\partial\bar{\omega}$:

$$Z^{(m)\alpha_1 \cdots \alpha_m} = \sum_r (\partial\bar{\omega})_{i_1} \cdots (\partial\bar{\omega})_{i_r} Z_{i_1 \cdots i_r}^{(m,r)\alpha_1 \cdots \alpha_m}, \quad (4.68)$$

where we may choose

$$A[i_1 \cdots i_r] Z_{i_1 \cdots i_r}^{(m,r)\alpha_1 \cdots \alpha_m} = Z^{(m,r)\alpha_1 \cdots \alpha_m}. \quad (4.69)$$

Here, A is the total antisymmetrizer of its arguments:

$$A = (1/r!) \sum_P \delta_P P,$$

$\delta_P = \pm 1$, according as P is an even or odd permutation of $(1, 2, \dots, r)$.

An application of the reasoning of Lemma A1 to Eq. (4.66), taking due account of the antisymmetry in $i_1, \dots, i_r$, leads to the decomposition

$$Z_{i_1 \cdots i_r}^{(m,r)\alpha_1 \cdots \alpha_m} = Z'_{i_1 \cdots i_r}{}^{\alpha_1 \cdots \alpha_m} + (A[\beta_1 \cdots \beta_r] J_{\beta_1 \cdots \beta_r}^\alpha) \partial_{i_1}^{\beta_1} \partial_{i_2}^{\beta_2} \cdots \partial_{i_r}^{\beta_r}, \quad (4.70)$$

with

$$Z'_{i_1 \cdots i_r}{}^{\alpha_1 \cdots \alpha_m} (D\Phi)_{i_1} = 0. \quad (4.71)$$

We may assume that Z' does not contain terms $\sim \partial_{i_s}$, $s = 1, \dots, r$. The reasoning of Lemma A1 tells us that

$$Z'_{i_1 \cdots i_r}{}^{\alpha_1 \cdots \alpha_m} = W_{i_1 \cdots i_r, j_1 \cdots j_p}^{(p)\alpha_1 \cdots \alpha_m} (D\Phi)_{j_1} \cdots (D\Phi)_{j_p}, \quad (4.72)$$

where

$$\bar{S}[j_1 \cdots j_p] W_{i_1 \cdots i_r, j_1 \cdots j_p}^{(p)\alpha_1 \cdots \alpha_m} = 0, \quad (4.73)$$

and

$$S[i_s, j_1 \cdots j_p] W_{i_1 \cdots i_r, j_1 \cdots j_p}^{(p)\alpha_1 \cdots \alpha_m} = 0, \quad 1 \leq s \leq r. \quad (4.74)$$

Therefore, $W^{(p)}$ is completely antisymmetric in $i_1, \dots, i_r$, symmetric in $i_1 \cdots j_p$, and satisfies the constraint, Eq. (4.74). We denote by e the identity element of the symmetric group S_{r+p} on $r+p$ objects, so that

$$e W_{i_1 \cdots i_r, j_1 \cdots j_p}^{(p)\alpha_1 \cdots \alpha_m} = W_{i_1 \cdots i_r, j_1 \cdots j_p}^{(p)\alpha_1 \cdots \alpha_m}. \quad (4.75)$$

The identity element e , has a resolution in terms of Young operators $Y_{\bar{c}}$ where Γ 's are standard Young tableaux [13]. We order the $r+p$ induces $i_1, \dots, i_r, j_1 \cdots j_p$ in this order. Then, the only Young operators that do not annihilate $W_{i_1 \cdots i_r, j_1 \cdots j_p}$ with the given symmetries, Eqs. (4.69), (4.73), and (4.74), correspond to the Young tableau Γ_s shown in Fig. 1. Therefore, we may write Eq. (4.75) as

$$W_{i_1 \cdots i_r, j_1 \cdots j_p} = \sum_{s=1}^p Y_{\Gamma_s} W_{i_1 \cdots i_r, j_1 \cdots j_p}, \quad (4.76)$$

FIG. 1. Young tableaux Γ_s , $s = 1, \dots, m$ whose Young operators do not annihilate $W_{i_1 \cdots i_r, j_1 \cdots j_p}$.

where the Y_{Γ} 's normalized to be idempotent:

$$Y_{\Gamma_s} Y_{\Gamma_r} = \delta_{s,r} Y_{\Gamma_s}. \quad (4.77)$$

Using Eq. (4.76) together with Eq. (4.72), we can write

$$\begin{aligned} \sum_r (\partial \bar{\omega})_{i_1} \cdots (\partial \bar{\omega})_{i_r} Z_{i_1 \cdots i_r}^{\alpha_1 \cdots \alpha_m} &= (D\Phi)_s \frac{\delta}{\delta(\partial \bar{\omega})_s} \left\{ \sum_{r,p} (\partial \bar{\omega})_{i_1} \cdots (\partial \bar{\omega})_{i_r} (\partial \bar{\omega})_{i_{r+1}} \right. \\ &\quad \times V_{i_1 \cdots i_r, l_r+2, k_1 \cdots k_{p-1}}^{(m,r,p)\alpha_1 \cdots \alpha_m} (D\Phi)_{k_1} \cdots (D\Phi)_{k_{p-2}} \left. \right\} \\ &\equiv (D\Phi)_s \frac{\delta}{\delta(\partial \bar{\omega})_s} V^{(m)\alpha_1 \cdots \alpha_m}[\Phi, \omega, \partial \bar{\omega}], \end{aligned} \quad (4.78)$$

where $V^{(m,r,p)}$ satisfies the symmetry conditions

$$\bar{A}[l_1 \cdots l_r l_{r+1}] V_{l_1 \cdots l_{r+1}, k_1 \cdots k_{p-1}}^{(m,r,p)} = 0, \quad (4.79)$$

$$\bar{S}[k_1 \cdots k_{p-2}] V_{l_1 \cdots l_{r+1}, k_1 \cdots k_{p-1}}^{(m,r,p)} = 0, \quad (4.80)$$

and in fact,

$$V_{i_1 \cdots i_r j_s j_1 \cdots j_p}^{(m,r,p)} = (p/(r+1)) Y_{\Gamma_s} W_{i_1 \cdots i_r, j_1 \cdots j_p}^{(p)}(-)^r. \quad (4.81)$$

Similarly, we may write

$$\begin{aligned} \sum_r (\partial \bar{\omega})_{i_1} \cdots (\partial \bar{\omega})_{i_r} (A[\beta_1 \cdots \beta_r] J_{\beta_1 \cdots \beta_r}^{\alpha_1 \cdots \alpha_m}) \partial_{i_1}^{\beta_1} \cdots \partial_{i_r}^{\beta_r} L_{\alpha_1} \cdots L_{\alpha_m} \\ = (\partial \bar{\omega})_i \partial_i^\alpha (\delta / \delta L_\alpha) U^{(m+1)}[\Phi, \omega, \partial \omega, L], \end{aligned} \quad (4.82)$$

where

$$\begin{aligned} U^{(m+1)} = \sum_{s=2} (\partial \bar{\omega})_{i_2} \cdots (\partial \bar{\omega})_{i_s} (1/(m+1)) (A[\beta \beta_2 \cdots \beta_s] J_{\beta \beta_2 \cdots \beta_s}^{\alpha_1 \cdots \alpha_m}) \\ \times \partial_{i_2}^{\beta_2} \cdots \partial_{i_s}^{\beta_s} L_{\alpha_1} \cdots L_{\alpha_m} L_\beta. \end{aligned} \quad (4.83)$$

Likewise, we define

$$V^{(m)}[\Phi, \omega, \partial \omega, L] = V^{(m)\alpha_1 \cdots \alpha_m}[\Phi, \omega, \partial \omega] L_{\alpha_1} \cdots L_{\alpha_m}. \quad (4.84)$$

What we have shown so far is that, given

$$(D\Phi)_i (\delta / \delta (\partial \bar{\omega})_i) Z^{(m)} = 0, \quad (4.85)$$

there exist $V^{(m)}$ and $U^{(m+1)}$, such that

$$Z^{(m)} = (D\Phi)_i (\delta / \delta (\partial \bar{\omega})_i) V^{(m)} + (\partial \bar{\omega})_i \partial_i^\alpha (\delta / \delta L_\alpha) U^{(m+1)}. \quad (4.86)$$

Further, note that

$$(D\Phi)_i (\delta / \delta (\partial \bar{\omega})_i) U^{(m+1)} = 0, \quad (4.87)$$

as evident from Eq. (4.83).

Now, assume that

$$\begin{aligned} Z^{(n)} = (D\Phi)_i (\delta / \delta (\partial \bar{\omega})_i) V^{(n)} + (\partial \bar{\omega})_i \partial_i^\alpha (\delta / \delta L_\alpha) [U^{(n+1)} + V^{(n+1)}], \\ (D\Phi)_i (\delta / \delta (\partial \bar{\omega})_i) U^{(n+1)} = 0_\eta \quad V^{(m+1)} \equiv 0. \end{aligned} \quad (4.88)$$

Then, by virtue of Eq. (4.65), we have

$$(D\Phi)_i (\delta/\delta(\partial\bar{\omega})_i) Z^{(n-1)} + (\partial\bar{\omega})_i \partial_i^\alpha (\delta/\delta L_\alpha) \{(D\Phi)_j (\delta/\delta(\partial\bar{\omega})_j) V^{(n)}\} = 0, \quad (4.89)$$

because

$$\{(\partial\bar{\omega})_i \partial_i^\alpha (\delta/\delta L_\alpha)\}^2 = 0.$$

Equation (4.89) can be written as

$$(D\Phi)_i (\delta/\delta(\partial\bar{\omega})_i) \{Z^{(n-1)} - (\partial\bar{\omega})_j \partial_j^\alpha (\delta/\delta L_\alpha) V^{(n)}\} = 0. \quad (4.90)$$

This is the same equation as Eq. (4.85). Therefore, there exist $V^{(n-1)}$ and $U^{(n)}$ such that

$$Z^{(n-1)} = (D\Phi)_i (\delta/\delta(\partial\bar{\omega})_i) V^{(n-1)} + (\partial\bar{\omega})_i \partial_i^\alpha (\delta/\delta L_\alpha) [V^{(n)} + U^{(n)}], \quad (4.91)$$

with

$$(D\Phi)_i (\delta/\delta(\partial\bar{\omega})) V^{(n)} = 0.$$

Thus, our inductive argument is complete and it terminates when $n - 1 = 0$ is reached. That is

$$Z = [(D\Phi)_i (\delta/\delta(\partial\bar{\omega})_i) + (\partial\bar{\omega})_i \partial_i^\alpha (\delta/\delta L_\alpha)] [V + U], \quad (4.92)$$

where

$$V = \sum_{n=0}^m (1/n!) V^{(n)},$$

$$U = \sum_{n=0}^m (1/n!) U^{(n+1)}.$$

This completes the proof of Eq. (4.60), and noting the arguments given between Eqs. (4.48) and (4.60), we conclude that all the solutions of the Eq. (4.3) containing equal numbers of ghosts and antighosts and at least one of each can be expressed as $\mathcal{G}'F[\Phi, \omega, \bar{\omega}, L]$. Since we had shown earlier that all other solutions with an equal number of c , $\bar{c}$'s can be expressed as, $\{\mathcal{G}'F[\Phi, \omega, \bar{\omega}, L]\} \oplus \{O^G I[\Phi]\}$, the Main Theorem is proved [14].

We may now expand $F[\Phi, \omega, \bar{\omega}, L]$ as

$$F[\Phi, \omega, \bar{\omega}, L] = F[\Phi, \omega, \bar{\omega}] - L_\alpha X_\alpha + O(L^2). \quad (4.93)$$

Then

$$\mathcal{G}'F[\Phi, \omega, \bar{\omega}, L]|_{L=0} \left(\mathcal{G}_0 + \frac{\delta \tilde{S}}{\delta \Phi_i} \frac{\delta}{\delta(\partial\bar{\omega})_i} \right) F[\Phi, \omega, \bar{\omega}] + \frac{\delta \tilde{S}}{\delta \omega_\alpha} X_\alpha[\Phi, \omega, \bar{\omega}]. \quad (4.94)$$

V. STRUCTURE OF THE RENORMALIZATION MATRIX

In this section, we shall prove another important result. It was shown in the last section that the complete set of gauge noninvariant operators that mix with a given set of gauge invariant operators can be expressed as

$$O[\Phi, \omega, \bar{\omega}] = \frac{\delta \tilde{S}}{\delta \Phi_i} \frac{\delta F}{\delta (\partial \bar{\omega})_i} + \frac{\delta \tilde{S}}{\delta \omega_\alpha} X_\alpha + \mathcal{G}_0 F, \quad (5.1)$$

where F and X_α are arbitrary local polynomials of appropriate ghost number and appropriate global and Lorentz transformation properties. It is clear that one can expand the divergence in $\Gamma_N[\Phi, \omega, \bar{\omega}]$ in the basis that consists of all independent linear combinations of the operators $\{O^{GI}[\Phi]\}' \oplus \{O[\Phi, \omega, \bar{\omega}]\}$. It is important to recognize the trivial fact that the matrix elements of the matrix Z , will depend on the basis one has chosen; and therefore, any statement about relationships of matrix elements of Z with its eigenvalues are bound to be basis dependent, in general.

In this section, we shall show that mixing with gauge noninvariant operators (and ghost operators) can be made irrelevant in the limited sense that there exists a basis of operators such that when the matrix Z is expressed in this basis, the diagonal submatrix Z^{GI} of Z yields, in fact, the correct eigenvalues corresponding to all eigenvectors of Z that involve gauge invariant operators. The basis is, in fact, $\{O^{GI}[\Phi]\}' \oplus$ gauge noninvariant operators of the form $O[\Phi, \omega, \bar{\omega}]$ of Eq. (5.1); and the diagonal submatrix Z^{GI} relates the divergences in the insertions of gauge invariant operators that are proportional to $\{O^{GI}[\Phi]\}'$ *when expanded in this basis*.

To show this, it is sufficient to show that the divergence in the insertion of a gauge noninvariant operator of the form $O[\Phi, \omega, \bar{\omega}]$ of Eq. (5.1) does not contain terms proportional to gauge invariant operators $\{O^{GI}\}'$ when expanded in the above basis.

We consider the generating functional of Green's functions

$$W[j, N] = \int [dA \, dc \, d\bar{c}] \exp i\{\tilde{S}[A, c, \bar{c}] - \frac{1}{2}\{f_\alpha[A]\}^2 + j_i A_i + NO[A, c, \bar{c}]\}, \quad (5.2)$$

$$O[A, c, \bar{c}] = \frac{\delta \tilde{S}}{\delta A_i} \frac{\delta F[A, c, \bar{c}]}{\delta (\partial \bar{c})_i} + \frac{\delta \tilde{S}}{\delta c_\alpha} X_\alpha[A, c, \bar{c}] + \mathcal{G}_0 F[A, c, \bar{c}]. \quad (5.3)$$

To see if gauge invariant operators appear in the divergence with a single insertion, it is sufficient to compute $\Gamma_N[\Phi, \omega = 0, \bar{\omega} = 0]$, i.e., it suffices to consider W as a function of j omitting the dependence on ghost sources. We perform an N -dependent transformation of the integration variables:

$$\begin{aligned} A_i &= A_i' - N \delta F[A, c, \bar{c}] / \delta (\partial \bar{c})_i, \\ c_\alpha &= c_\alpha' - N X_\alpha[A, c, \bar{c}], \quad \bar{c} = \bar{c}'. \end{aligned} \quad (5.4)$$

We also note that F and X_α are local functionals and the Jacobian J of these transformation is

$$\ln J = N \left(\frac{\delta F}{\delta A_i \delta(\partial \bar{c})_i} - \frac{\delta X_\alpha}{\delta c_\alpha} \right),$$

and $\ln J$ is proportional to $\delta^4(0)$ or derivatives of $\delta^4(x)$ at $x = 0$. We shall use dimensional regularization, wherein $\delta^4(0)$, or are interpreted to be zero. Thus, $W[j, N]$ is given by

$$\begin{aligned} W[j, N] = & \int [dA' dc' d\bar{c}'] \exp i \left\{ \tilde{S}[A', c', \bar{c}'] - \frac{1}{2} \eta_0 \left[\partial_i^\alpha \left[A_i' - N \frac{\delta F}{\delta(\partial \bar{c})_i} \right] \right]^2 \right. \\ & \left. + J_i A_i + \mathcal{G}_0 F - J_i N \frac{\delta F}{\delta(\partial \bar{c})_i} \right\} + O(N^2). \end{aligned} \quad (5.5)$$

Omitting primes, we obtain

$$\frac{1}{W[j]} \frac{\delta W[j, N]}{\delta N} \Big|_{N=0} \equiv \langle O \rangle_j \quad (5.6)$$

$$\begin{aligned} &= \int [dA dc d\bar{c}] \left\{ -j_i \frac{\delta F}{\delta(\partial \bar{c})_i} + \eta_0 \partial_i^\alpha A_i \partial_i^\alpha \frac{\delta F}{\delta(\partial \bar{c})_i} + \mathcal{G}_0 F \right\} \exp i \{ \mathcal{L}_{\text{eff}} + j_i A_i \} \\ &\equiv -j_i \left\langle \frac{\delta F}{\delta(\partial \bar{c})_i} \right\rangle_j + \eta_0 \left\langle \partial_j^\alpha A_j \partial_i^\alpha \frac{\delta F}{\delta(\partial \bar{c})_i} + \mathcal{G}_0 F \right\rangle_j. \end{aligned} \quad (5.7)$$

Next, consider

$$\langle F \rangle_j = \int [dA dc d\bar{c}] F[A, c, \bar{c}] \exp i \{ \mathcal{L}_{\text{eff}}[A, c, \bar{c}] + j_i A_i \}. \quad (5.8)$$

We perform the BRS transformations on the integration variables given in Eq. (2.6) and equate the total change to zero. Noting that the Jacobian of the transformation is one, we obtain.

$$0 = \int [dA dc d\bar{c}] \left\{ \mathcal{G}_0 F + \frac{\delta F}{\delta(\partial \bar{c})_i} \eta_0 \partial_j^\alpha A_j \partial_i^\alpha - ij_i D_i^\beta c_\beta F \right\} \exp i \{ \mathcal{L}_{\text{eff}} + j_i A_i \}, \quad (5.9)$$

i.e.

$$\left\langle \mathcal{G}_0 F + \eta_0 \partial_i^\alpha \partial_j^\alpha A_j \frac{\delta F}{\delta(\partial \bar{c})_i} \right\rangle_j = ij_i \langle D_i^\beta c_\beta F \rangle_j. \quad (5.10)$$

Using Eq. (5.10) in Eq. (5.7), we get

$$\langle O \rangle_j = -j_i \left\langle \frac{\delta F}{\delta(\partial \bar{c})_i} \right\rangle_j + ij_i \langle D_i^\beta c_\beta F \rangle_j. \quad (5.11)$$

We write

$$-j_i[\phi] = \frac{\delta \Gamma_0}{\delta \phi_i} = \frac{\delta \Gamma_0[\phi]}{\delta \phi_i} - \eta_0 \partial_i^\alpha \partial_j^\alpha \phi_j, \quad (5.12)$$

and thus obtain,

$$\langle O \rangle_j = \frac{\delta \Gamma_0[\phi]}{\delta \phi_i} \left\{ \left\langle \frac{\delta F}{\delta(\partial \bar{c})_i} \right\rangle_j - i \langle D_i^\alpha c_\alpha F \rangle_j \right\} - \eta_0 \partial_j^\alpha \phi_j \left\langle \partial_i^\alpha \frac{\delta F}{\delta(\partial \bar{c})_i} - \partial_i^\alpha D_i^\beta c_\beta F \right\rangle_j. \quad (5.13)$$

The last term vanishes since

$$\begin{aligned} \left\langle \partial_i^\alpha \frac{\delta F}{\delta(\partial \bar{c})_i} - \partial_i^\alpha D_i^\beta c_\beta F \right\rangle_j &= \left\langle \frac{\delta F}{\delta \bar{c}_\alpha} - i(Mc)_\beta F \right\rangle_j \\ &= \int [dA] e^{i(\mathcal{L}_0 - 1/2 f_\alpha^2 + j_i A_i)} \int [dc \, d\bar{c}] \frac{\delta}{\delta \bar{c}_\alpha} [F e^{i\bar{c} M c}] = 0. \end{aligned} \quad (5.14)$$

Therefore,

$$\langle O \rangle_j = \frac{\delta \Gamma_0[\phi]}{\delta \phi_i} \left\langle \frac{\delta F}{\delta(\partial \bar{c})_i} - i D_i^\alpha c_\alpha F \right\rangle_j, \quad (5.15)$$

$$= \frac{\delta \Gamma_0^{(r)}[\Phi]}{\delta \Phi_i} \left\{ Z^{-1/2} \left\langle \frac{\delta F}{\delta(\partial \bar{c})_i} - i D_i^\alpha c_\alpha F \right\rangle_j \right\} \quad (5.16)$$

$$\equiv \frac{\delta \Gamma_0^{(r)}[\Phi]}{\delta \Phi_i} \mathcal{F}_i[\phi]. \quad (5.17)$$

$\mathcal{F}_i[\phi]$ is to be renormalized in each loop approximation $\{\mathcal{F}_i[\phi]\}_n^{\text{div}}$, the overall divergence in $\mathcal{F}_i[\phi]$ expressed in terms of renormalized fields and parameters, *always* can be expressed in terms of a complete set of functionals with the same Lorentz and global transformation properties and dimensions as $\mathcal{F}_i[\phi]$. Let this set be $\{F_i^{(p)}[\Phi]\}$. Then

$$\{\mathcal{F}_i[\phi]\}_n^{\text{div}} = \sum_p A_p^{(n)} F_i^{(p)}[\Phi]. \quad (5.18)$$

Thus

$$\{\langle O \rangle_j\}_n^{\text{div}} = \frac{\delta \mathcal{L}_0[\Phi]}{\delta \Phi_i} \sum_p A_p^{(n)} F_i^{(p)}[\Phi]. \quad (5.19)$$

But $\delta \mathcal{L}_0 / \delta \Phi_i F_i^{(p)}[\Phi]$ are precisely the ghost independent terms of the gauge non-invariant operators in our basis: $\mathcal{G}' F_i^{(p)}(\partial \bar{c})_i$. It should be noted that, in the set of gauge noninvariant operators $\mathcal{G}' F$, F runs over *all* functionals of the appropriate kind. Therefore, $\{\langle O \rangle_j\}_n^{\text{div}}[\Phi, \omega, \partial \bar{\omega}]$ is expressible as a linear combination of O 's in Eq. (5.1), not including O_{GI} 's.

We further note that in Eq. (5.1), functionals $F[\Phi, \omega, \bar{\omega}]$ and $X_\alpha[\Phi, \omega, \bar{\omega}]$ are uncorrelated, so that it is convenient to separate the basis into two parts.

$$\begin{aligned} \text{class I operators: } & \frac{\delta \tilde{S}}{\delta \Phi_i} \frac{\delta F}{\delta (\partial \bar{\omega})_i} + \mathcal{G}_0 F \\ \text{class II GNI operators: } & \frac{\delta \tilde{S}}{\delta \omega_\alpha} X_\alpha[\Phi, \omega, \bar{\omega}]. \end{aligned} \quad (5.20)$$

(Class I operators, in addition to *GNI* operators, contains *GI* operators that have the form $\delta \mathcal{L}_0 / \delta A_i F_i[A]$.) The divergence in the single insertion of a class II operator can be expanded in terms of class II operators only. To see this, consider the generating functional:

$$\begin{aligned} W[j, \xi, \bar{\xi}, N] = \int [dA \, dc \, d\bar{c}] \exp i \left\{ \tilde{S}[A, c, \bar{c}] - \frac{1}{2} \{f_\alpha\}^2 + j_i A_i + N \frac{\delta \tilde{S}}{\delta c_\alpha} X_\alpha \right\} \\ + \bar{\xi}_\alpha c_\alpha + \bar{c}_\alpha \xi_\alpha. \end{aligned} \quad (5.21)$$

By performing a transformation on c_α only

$$c_\alpha \rightarrow c_\alpha + N X_\alpha, \quad (5.22)$$

and noting that the change in the Jacobian is zero [$X_{\alpha,\alpha} \sim \delta^4(0)$] in dimensional regularization. We obtain (dropping primes)

$$\begin{aligned} W[j, \xi, \bar{\xi}, N] = \int [dA \, dc \, d\bar{c}] \exp i \left\{ \tilde{S}[A, c, \bar{c}] - \frac{1}{2} f_\alpha^2 + j_i A_i + c_\alpha \xi_\alpha + \bar{\xi}_\alpha c_\alpha + \bar{\xi}_\alpha N X_\alpha \right\} \\ + O(N^2) \text{ terms.} \end{aligned} \quad (5.23)$$

Thus,

$$\begin{aligned} \left(\frac{1}{W} \frac{\delta W}{\delta N} \right)_{N=0} &= \bar{\xi}_\alpha \langle X_\alpha \rangle_{j, \xi, \bar{\xi}} \\ &= \frac{\delta \Gamma}{\delta \Omega_\alpha} \langle X_\alpha \rangle_{j, \xi, \bar{\xi}} = \frac{\delta \Gamma^{(r)}[\Phi, \omega, \bar{\omega}]}{\delta \omega_\alpha} \langle \tilde{Z}^{-1/2} X_\alpha \rangle_{j \xi \bar{\xi}}, \end{aligned} \quad (5.24)$$

$\langle X_\alpha \rangle_{j, \xi, \bar{\xi}}$ is a functional of $\Phi, \omega, \bar{\omega}$. It is to be renormalized as we renormalized F . (See the discussion following Eq. (5.17).) Thus,

$$\begin{aligned} \{\Gamma_N[\Phi, \omega, \bar{\omega}]\}_n^{\text{div}} &= \left\{ \frac{1}{W} \frac{\delta W[j, \xi, \bar{\xi}, N]}{\delta N} \right\}_{N=0}_n^{\text{div}} \\ &= \frac{\delta \tilde{S}[\Phi, \omega, \bar{\omega}]}{\delta \omega_\alpha} \{ \langle \tilde{Z}^{-1/2} X_\alpha \rangle \}_n^{\text{div}}, \end{aligned} \quad (5.25)$$

and is expressible as the sum of class II *GNI* operators only.

We have thus shown that the matrix Z has the structure:

$$Z = \begin{array}{c} \begin{array}{cc} & \begin{array}{cc} GI' & \text{I} & \text{II} \end{array} \\ \begin{array}{c} GI' \\ \text{I} \\ \text{II} \end{array} & \left[\begin{array}{c|c|c} Z^{GI'} & & \\ \hline 0 & & \\ \hline 0 & 0 & \end{array} \right] \end{array} . \quad (5.26)$$

It is obvious that eigenvalues of $Z^{GI'}$ are also the eigenvalues of Z . Furthermore, the eigenstates of Z corresponding to these eigenvalues are the only eigenstates that involve a gauge invariant operators $\{O^{GI'}\}$ in this basis. Thus, the eigenvalues corresponding to the eigenstates of Z that involve a gauge invariant operator in this basis can be obtained only by computing $Z^{GI'}$. One cannot, however, compute these eigenstates simply by knowing $Z^{GI'}$. In practice, $Z^{GI'}$ can be computed by finding some distinguishing property of $\{O^{GI'}[\Phi]\}'$ that sets it apart from the gauge noninvariant operators with which it mixes. For example, in the case of the twist two operators, only the 2-point function of $O^{GI'}[\Phi]$ [F.T. $(\delta O^{GI'}[\Phi]/\delta \Phi_\mu^\alpha \delta \Phi_\nu^\alpha)$] contains terms proportional to $g_{\mu\nu}$, while the GNI operators with which it mixes do not.

APPENDIX A.

In this Appendix, we wish to show that a local solution of the equation

$$\mathcal{L}_{0,i}[\Phi] P_i^\alpha[\Phi] = 0, \quad (A1)$$

has a structure discussed in Lemma A2 below. We begin with a series of definitions and a lemma.

We define

$$D_{ij} = \delta_{\alpha\beta}(\partial^2 g_{\mu\nu} - \partial_\mu \partial_\nu) \delta^4(x - y), \quad (A2)$$

where $i = (\alpha, \mu, x)$ and $j = (\beta, \nu, y)$. Note that

$$\mathcal{L}_{0,ij}[\Phi = 0] = -D_{ij}. \quad (A3)$$

We define the transverse and longitudinal projection operators, T_{ij} and L_{ij} by

$$D_{ij} = \partial_i^2 T_{ij}, \quad \delta_{ij} = T_{ij} + L_{ij}, \quad (A4)$$

which satisfy

$$T_{ik} T_{kj} = T_{ij}, \quad L_{ik} L_{kj} = L_{ij}, \quad (A5)$$

$$T_{ik} L_{kj} = L_{ik} T_{kj} = 0. \quad (A6)$$

We consider first a simpler case.

LEMMA A1. Any functional $P_i^\alpha[\Phi]$ that contains at most n fields, and that satisfies

$$P_i^\alpha[\Phi] D_{ij} \Phi_j = 0, \quad (\text{A7})$$

can be represented as

$$P_i^\alpha[\Phi] = Z^{\alpha\beta}[\Phi] \partial_i^\beta + V_{ij}^\alpha[\Phi] D_{jk} \Phi_k, \quad (\text{A8})$$

where

$$V_{i_1 i_2}^\alpha[\Phi] = \sum_{r=1}^n V_{i_1 i_2 \dots i_{r+1}}^{\alpha(r)}[\Phi] \prod_{s=3}^{r+1} (D_{i_s k} \Phi_k), \quad (\text{A9})$$

with

$$\sum_{\{i\}} V_{i_1 i_2 \dots i_{r+2}}^{\alpha(r)} = 0. \quad (\text{A10})$$

The symbol $\sum_{\{i\}}$ denotes summation over permutations of the $r+2$ indices $\{i_1, i_2, \dots, i_{r+2}\}$.

Proof of A1. It suffices to consider a monomial

$$P_i^\alpha[\Phi] = P_{i, i_1 \dots i_n}^\alpha \Phi_{i_1} \Phi_{i_2} \dots \Phi_{i_n}. \quad (\text{A11})$$

The kernel $P_{i, i_1 \dots i_n}^\alpha$ is completely symmetric in the n indices $\{i_1, \dots, i_n\}$. Equation (A7) then implies

$$\sum_{\{i_1, \dots, i_{n+1}\}} P_{i, i_1 \dots i_n}^\alpha D_{i, i_{n+1}} = 0, \quad (\text{A12})$$

where the summation is over permutations of the $n+1$ indices $\{i_1, \dots, i_{n+1}\}$.

We can decompose $P_{i, i_1 \dots i_n}^\alpha$ with respect to each of its indices using the partition of identity (A4). For example, we write

$$P_{i, i_1 \dots i_n}^\alpha = L_{ik} P_{k, i_1 \dots i_n}^\alpha + T_{ik} P_{k, i_1 \dots i_n}^\alpha. \quad (\text{A13})$$

The first term on the right does not contribute at all in Eq. (A12), and yields the first term on the right of Eq. (A8), $Z^{\alpha\beta} \partial_i^\beta$ with a suitably defined functional $Z^{\alpha\beta}$. From now on, we shall assume that $P_{i, i_1 \dots i_n}^\alpha$ is transverse with respect to i .

We shall write

$$P_{i, i_1 \dots i_n}^\alpha = \sum_{s=0}^n \sum_{\{i_1, \dots, i_n\}} P_{i, i_1 \dots i_s; i_{s+1} \dots i_n}^{\alpha(s)}, \quad (\text{A14})$$

where $P_{i, i_1 \dots i_s; i_{s+1} \dots i_n}^{\alpha(s)}$ is transverse with respect to i_j , $1 \leq j \leq s$, and is longitudinal with respect to i_j , $j \geq s+1$. Substituting Eq. (A14) in Eq. (A12) and setting

$$P_{j, i_1 \dots i_s; i_{s+1} \dots i_n}^{\alpha(s)} = V_{j j_1 j_2 \dots j_s; \beta_{s+1} \dots \beta_n}^{\alpha(s)} D_{ji} D_{j_1 i_1} \dots D_{j_s i_s} \partial_{i_{s+1}}^{\beta_{s+1}} \dots \partial_{i_n}^{\beta_n}, \quad (\text{A15})$$

we obtain

$$\sum_{\{i_1, \dots, i_s\}} V_{jj_1 j_2 \dots j_s; \beta_{s+1} \dots \beta_n}^{\alpha(s)} D_{ji} D_{j_1 i_1} \dots D_{j_s i_s} \partial_{i_{s+1}}^{\beta_{s+1}} \dots \partial_{i_n}^{\beta_n} = 0. \quad (\text{A16})$$

For $s = 0$, we must have

$$V_{j; \beta_1 \dots \beta_n}^{\alpha(0)} = 0. \quad (\text{A17})$$

For $s \neq 0$, we may set, without loss of generality,

$$\sum_{\{j_1, \dots, j_s\}} V_{jj_1 \dots j_s; \beta_{s+1} \dots \beta_n}^{\alpha(s)} = 0. \quad (\text{A18})$$

The structure implied by Eqs. (A9) and (A10) follows from (A17) and (A18). This completes the proof of Lemma A1.

For later use, we note the special case when $P_i^\alpha[\Phi]$ is restricted to be a local functional. This means that

$$P_{i, i_1 \dots i_n}^\alpha = f(\partial_j) \delta^4(x_1 - x_\alpha) \delta^4(x_{i_1} - x_\alpha) \dots \delta^4(x_{i_n} - x_\alpha), \quad (\text{A19})$$

where $f(\partial_j)$ is a polynomial in differential operators. In general, a functional of the transverse and longitudinal components of the vector field Φ_j :

$$\Phi_j = \Phi_j^T + \partial_j^\alpha \Phi_\alpha^L,$$

or

$$\Phi_j^T = \left(\delta_{ji} - \partial_j \frac{1}{\partial^2} \partial_i \right) \Phi_i, \quad (\text{A20})$$

$$\Phi_\alpha^L \sim \partial_i^\alpha \frac{1}{\partial^2} \partial_j^\alpha \Phi_j,$$

is not a local unless Φ_j^T and Φ_α^L enter the expression (i) through the combination $\Phi_j^T + \partial_j^\alpha \Phi_\alpha^L$, or (ii) in the form $\partial^2 \Phi_j^T = D_{ji} \Phi_i$ and $\partial^2 \Phi_\alpha^L = \partial_i^\alpha \Phi_i$.

Consider the term in Eq. (A14), which contains the highest power of Φ_i^T , i.e., the term with $s = n$. In this case, Eq. (A18) reads

$$\sum_{\{j, j_1 j_2 \dots j_n\}} V_{jj_1 \dots j_n}^{\alpha(n)} = 0 \quad (\text{A21})$$

Suppose now $V_{ii_1 \dots i_n}^{\alpha(n)}$ has a factor $\partial_{j_1}^{-2} \partial_{j_2}^{-2} \dots \partial_{j_n}^{-2}$ (note that $V_{jj_1 \dots j_n}$ must be symmetric in $j_1, \dots, j_n$). Then, Eq. (A21) tells us that $V_{jj_1 \dots j_n}$ must have a factor ∂_j^{-2} , and consequently, from Eq. (A15) $P_{j, i_1 \dots i_n}^{\alpha(n)}$ is not local. Therefore, $V_{jj_1 \dots j_n}^{\alpha(n)}$ must be local. Repeating this argument, and taking into account the points (i) and (ii) above, we find that for a local functional satisfying (A7), $V_{i_1 \dots i_{r+1}}^{\alpha(r)}$ must be a local functional of $\partial_i^\alpha \Phi_i$.

We are now prepared to prove the Main Lemma.

LEMMA A2. *Any local functional $P_i^\alpha(\Phi)$ of Φ_j that satisfies:*

$$P_i^\alpha[\Phi] \mathcal{L}_{0,i}[\Phi] = 0, \quad (\text{A22})$$

can be represented as

$$P_i^\alpha[\Phi] = X^{\alpha\beta}[\Phi] D_i^\beta[\Phi] + Y_{ij}^\alpha[\Phi] \mathcal{L}_{0,j}[\Phi], \quad (\text{A23})$$

where $X^{\alpha\beta}$ and Y_{ij}^α are suitable functionals of Φ_k .

Since $P_i^\alpha[\Phi]$ is restricted to a local functional, we can meaningfully discuss its dimension. Since $\mathcal{L}_{0,i}$ has a definite dimension, we may assume, without loss of generality, that $P_i^\alpha[\Phi]$ has a definite dimension N . We prove this lemma by means of the previous one.

Proof of A2. We expand $P_i^\alpha[\Phi]$ as

$$P_i^\alpha[\Phi] = \sum_{n=0}^N P_{ij_1 \dots j_n}^\alpha \Phi_{j_1} \cdots \Phi_{j_n}, \quad (\text{A24})$$

and substitute it in Eq. (A22). We equate the coefficient of lowest power of Φ to zero and obtain

$$P_i^\alpha[\Phi = 0] D_{ij} \Phi_j = 0. \quad (\text{A25})$$

By the previous lemma, we have

$$P_i^\alpha[\Phi = 0] = X^{\alpha\beta} \partial_i^\beta. \quad (\text{A26})$$

We now define

$$P_i^{\alpha(1)}[\Phi] = P_i^\alpha[\Phi] - X^{\alpha\beta} D_i^\beta[\Phi]. \quad (\text{A27})$$

$P_i^{\alpha(1)}$ is a local functional with dimension N , begins with first power of Φ , and satisfies

$$P_i^{\alpha(1)}[\Phi] \mathcal{L}_{0,i}[\Phi] = 0. \quad (\text{A28})$$

Obviously, this process can be continued. Assume $P_i^{\alpha(n)}$ is a local functional of dimension N , begins with Φ^n , and satisfies

$$P_i^{\alpha(n)}[\Phi] \mathcal{L}_{0,i}[\Phi] = 0. \quad (\text{A29})$$

Let the Φ^n term of $P_i^{\alpha(n)}$ be $[P_i^{\alpha(n)}]$. Then,

$$[P_i^{\alpha(n)}] D_{ij} \Phi_j = 0. \quad (\text{A30})$$

Then, by Lemma A1, we have

$$[P_i^{\alpha(n)}] = X^{\alpha\beta(n)}[\Phi] \partial_i^\beta + \sum_{r=1}^n V_{ij_1 \dots j_r}^{\alpha(n,r)} [\partial_k^\beta \Phi_k] \prod_{s=1}^r (D_{j_s l} \Phi_l), \quad (\text{A31})$$

where $X^{\alpha\beta(n)}$ and $V^{\alpha(n,r)}$ are n th and $(n-r)$ th order functionals of Φ of dimension $N-1$ and $N-3r$, respectively. We then construct

$$P_i^{\alpha(n+1)}[\Phi] = P_i^{\alpha(n)}[\Phi] - X^{\alpha\beta(n)}[\Phi] D_i^\beta[\Phi] - \sum_{r=1}^n V_{ij_1 \dots j_r}^{\alpha(n,r)} [\partial \Phi] \prod_{s=1}^r \mathcal{L}_{0,j_s}[\Phi] \quad (\text{A32})$$

$$P_i^{\alpha(n+1)}[\Phi] \mathcal{L}_{0,i}[\Phi] = 0.$$

which is a local functional of dimension N and begins with Φ^{n+1} .

This process must terminate at some step, because the only solution to Eq. (A30) for a fixed dimension N and large enough n is zero. Suppose that $[P_i^{\alpha(n)}] = 0$ for $n \geq m$. Then, $P_i^{\alpha(m)} = 0$.

We can solve the system of equations [A32] for $n = 0, 1, \dots, m-1$ for $P_i^\alpha = P_i^{\alpha(0)}$ and find that it has the form

$$P_i^\alpha[\Phi] = X^{\alpha\beta}[\Phi] D_i^\beta[\Phi] + Y_{ij}^\alpha[\partial \Phi] \mathcal{L}_{0,j}[\Phi].$$

APPENDIX B

DEFINITIONS. We define the symmetrizer $S[i_1 \dots i_p]$ by

$$S[i_1 \dots i_p] f_{i_1 \dots i_p A} = \frac{1}{p!} \sum_{\{i_1 \dots i_p\}} f_{i_1 \dots i_p A}, \quad (\text{B1})$$

where the summation is over permutations of the p indices $\{i_1 \dots i_p\}$ and A denotes collectively other indices. We define the complement $\bar{S}[i_1 \dots i_p]$ of $S[i_1 \dots i_p]$ by

$$\bar{S}[i_1 \dots i_p] + S[i_1 \dots i_p] = 1. \quad (\text{B2})$$

The symmetrizer and its complement are projection operators and commute with $\mathcal{G}$ of Eq. (4.28)

$$S^2 = S, \quad \bar{S}^2 = \bar{S}, \quad (\text{B3})$$

$$S\bar{S} = 0 = \bar{S}S,$$

$$[\mathcal{G}, S] = 0 = [\mathcal{G}, \bar{S}]. \quad (\text{B4})$$

LEMMA B1. *All solutions of*

$$\sum_{r=1}^n C_{\eta i_1 \dots i_r \dots i_n}^{\gamma} [\Phi] D_{i_r}^{\gamma} = 0 \quad (\text{B5})$$

can be expressed as

$$C_{\eta i_1 \dots i_r \dots i_n}^{\gamma_1} [\Phi] = \sum_{t=2}^{n-1} E_{\eta i_1 \dots i_r \dots i_t \dots i_n}^{(\gamma_1 \dots \gamma_t)} [\Phi] D_{i_p}^{\gamma_2} \dots D_{i_q}^{\gamma_t} \quad (\text{B6})$$

with

$$S[\gamma_1 \dots \gamma_t] E^{(\gamma_1 \dots \gamma_t)} = 0. \quad (\text{B7})$$

(A hat denotes the absence of that index.)

Proof of B1. The proof proceeds much the same way as the Lemma A2 in Appendix A. We can always assume that C^{γ} has a definite dimensions. We assume that $C_{i_1 \dots i_r \dots i_n}^{\gamma}$ begins with a term $\sim \Phi^m$. We denote the Φ^m terms in C^{γ} by $C^{\gamma(m)}$. Equating the Φ^m terms in Eq. (B5) to zero we obtain

$$\sum_r C_{\eta i_1 \dots i_r \dots i_n}^{\gamma(m)} [\Phi] \partial_{i_r}^{\gamma} = 0. \quad (\text{B8})$$

Now, $C_{\eta i_1 \dots i_r \dots i_n, \{j\}}^{\gamma(m)} \equiv C_{\eta i_1 \dots i_r \dots i_n, j_1 \dots j_m}^{\gamma(m)}$ can be expanded in powers of momenta associated with $i_1 \dots i_n$, which are independent. We write

$$C_{\eta i_1 \dots i_r \dots i_n}^{\gamma(m)} = \bar{C}_{\eta i_1 \dots i_r \dots i_n}^{\gamma(m)} + \sum_p \partial_{i_p}^{\alpha} C_{i_1 \dots i_r \dots i_n}^{\gamma \alpha(m)} + \sum_{pq} \partial_{i_p}^{\alpha} \partial_{i_q}^{\beta} C_{i_1 \dots i_r \dots i_n}^{\gamma \alpha \beta(m)} + \dots \quad (\text{B9})$$

Thus, Eq. (B8) implies

$$\bar{C}_{\eta i_1 \dots i_r \dots i_n}^{\gamma(m)} = 0 \quad (\text{B10})$$

$$S[\gamma, \alpha] C_{\eta i_1 \dots i_r \dots i_n}^{\gamma \alpha(m)} = 0. \quad (\text{B11})$$

$$S[\gamma_1 \dots \gamma_t] C_{\eta i_1 \dots i_r \dots i_n}^{\gamma_1 \dots \gamma_t} = 0, \quad \text{etc.} \quad (\text{B12})$$

We then form

$$C_{\eta i_1 \dots i_r \dots i_n}^{\gamma[m+1]} = C_{\eta i_1 \dots i_r \dots i_n}^{\gamma} - \left[\sum_p D_{i_p}^{\alpha} C_{\eta i_1 \dots i_r \dots i_n}^{\gamma \alpha(m)} + \sum_{p,q} D_{i_p}^{\alpha} D_{i_q}^{\beta} C_{\eta i_1 \dots i_r \dots i_n}^{\gamma \alpha \beta(m)} + \dots \right]. \quad (\text{B13})$$

Thus, $C^{\gamma[m+1]}$ of Eq. (B13) satisfies Eq. (B5) and begins as terms Φ^{m+1} . This process can be continued, in each step increasing the power of Φ in C^{γ} by one.

The process will terminate when the power of Φ exceeds the dimension of C^γ . Then, Eq. (B13) together with its analogs imply the structure of Eq. (B6) for C^γ .

LEMMA B2. *Suppose a local functional $P_i[\Phi, \omega]$ satisfies*

$$\mathcal{G}(P_i[\Phi, \omega]) = \omega_\alpha X^\alpha[\Phi, \omega] D_i^\gamma[\Phi], \quad (\text{B14})$$

where P_i may carry additional indices. Then there exists a local functional $Z^\beta[\Phi, \omega]$ such that

$$\mathcal{G}(P_i[\Phi, \omega] - Z^\beta[\Phi, \omega] D_i^\beta[\Phi]) = 0. \quad (\text{B15})$$

Proof of B2. Equation (B14) implies:

$$\mathcal{G}(\omega_\alpha X^{\alpha\gamma}[\Phi, \omega]) D_i^\gamma = 0. \quad (\text{B16})$$

We multiply Eq. (B11) by ∂_i^σ . Since $\partial_i^\sigma D_i^\gamma = M^{\sigma\gamma}[\Phi]$ is invertible, we have

$$\mathcal{G}(\omega_\alpha X^{\alpha\gamma}) = 0. \quad (\text{B17})$$

We can now apply the Dixon–Taylor lemma to Eq. (B17). This, together with the discussion of locality in Appendix C, implies

$$\omega_\alpha X^{\alpha\gamma} = \mathcal{G}(Z^\gamma), \quad (\text{B18})$$

for some local $Z^\gamma[\Phi, \omega]$. Now, since $\mathcal{G}(D_i^\gamma) = 0$,

$$\mathcal{G}[Z^\beta[\Phi, \omega] D_i^\beta[\Phi]] = \omega_\alpha X^{\alpha\gamma} D_i^\gamma \quad (\text{B19})$$

Q.E.D.

LEMMA B3. *Given the equation*

$$\mathcal{G} \left[\sum_{r=1}^n C_{i_1 \dots i_r \dots i_n}^\gamma[\Phi, \omega] D_{i_r}^\gamma \right] = 0, \quad (\text{B20})$$

we can always redefine $C_{i_1 \dots i_r \dots i_n}^\gamma$ without altering $[\sum_{r=1}^n C_{i_1 \dots i_r \dots i_n}^\gamma D_{i_r}^\gamma]$ such that

$$\mathcal{G}[C_{i_1 \dots i_r \dots i_n}^\gamma[\Phi, \omega]] = 0, \quad (\text{B21})$$

where the C 's are local functionals and may carry additional indices.

Proof of B3. Eq. (B20) implies

$$\sum_{r=1}^n \mathcal{G}(C_{i_1 \dots i_r \dots i_n}^\gamma) D_{i_r}^\gamma = 0. \quad (\text{B22})$$

Lemma B1 as applied to Eq. (B22) implies

$$\mathcal{G}(C_{i_1 \dots i_r \dots i_n}^{\gamma_1 \dots \gamma_t}) = \sum_{t=2} E_{i_1 \dots i_r i_p \dots i_q \dots i_n}^{\gamma_1 \dots \gamma_t} \underbrace{D_{i_p}^{\gamma_2} \dots D_{i_q}^{\gamma_t}}_t, \quad (\text{B23})$$

with

$$S[\gamma_1 \dots \gamma_t] E^{\gamma_1 \dots \gamma_t} = 0. \quad (\text{B24})$$

Henceforth, we suppress suffixes whenever inessential.

We define

$$E'^{\gamma_1 \dots \gamma_t} = S[\gamma_1 \dots \gamma_{t-1}] \bar{S}[\gamma_1 \dots \gamma_t] \sum_{p=0}^{n-t} E^{\gamma_1 \dots \gamma_{t+p}} D^{\gamma_{t+1}} \dots D^{\gamma_{t+p}} \quad (\text{B25})$$

with

$$S[\gamma] = 1.$$

Then, $E'^{\gamma_1 \dots \gamma_t}$ satisfies

$$\bar{S}[\gamma_1 \dots \gamma_r] E'^{\gamma_1 \dots \gamma_t} = 0, \quad 2 \leq r < t, \quad t > 2, \quad (\text{B26})$$

because

$$\bar{S}[\gamma_1 \dots \gamma_r] S[\gamma_1 \dots \gamma_t] = S[\gamma_1 \dots \gamma_t] \bar{S}[\gamma_1 \dots \gamma_r] = 0, \quad \text{for } r < t. \quad (\text{B27})$$

Further, using

$$\sum_{r=2}^n S[\gamma_1 \dots \gamma_{r-1}] \bar{S}[\gamma_1 \dots \gamma_r] = \bar{S}[\gamma_1 \dots \gamma_n], \quad (\text{B28})$$

and Eq. (B24), we find that

$$\sum_{t=2}^n E'^{\gamma_1 \dots \gamma_t} D^{\gamma_2} \dots D^{\gamma_t} = \sum_{t=2}^n E^{\gamma_1 \dots \gamma_t} D^{\gamma_2} \dots D^{\gamma_t}. \quad (\text{B29})$$

Thus, Eq. (B23) implies

$$\mathcal{G}(C_{i_1 \dots i_r \dots i_n}^{\gamma_1 \dots \gamma_t}) = \left(\sum_{t=2}^n E'^{\gamma_1 \gamma_2 \dots \gamma_t} D^{\gamma_3} \dots D^{\gamma_t} \right) D^{\gamma_2}, \quad (\text{B30})$$

or

$$\bar{S}[\gamma_1 \gamma_2] \mathcal{G} \sum_{t=2}^n (E'^{\gamma_1 \gamma_2 \dots \gamma_t} D^{\gamma_3} \dots D^{\gamma_t}) = \mathcal{G}(E'^{\gamma_1 \gamma_2}) = 0, \quad (\text{B31})$$

where use has been made of Eq. (B26). Repeating this process we find that

$$\mathcal{G}(E'^{\gamma_1 \cdots \gamma_t}) = 0, \quad 2 \leq t \leq n. \quad (\text{B32})$$

By the Dixon–Taylor lemma, therefore, there exist $F^{\gamma_1 \cdots \gamma_t}$, such that

$$E'^{\gamma_1 \cdots \gamma_t} = \mathcal{G}F^{\gamma_1 \cdots \gamma_t}, \quad (\text{B33})$$

with

$$S[\gamma_1 \cdots \gamma_t] F^{\gamma_1 \cdots \gamma_t} = 0. \quad (\text{B34})$$

We can write Eq. (B23) as

$$\mathcal{G}[C'_{i_1 \cdots i_r \cdots i_n}] = 0 \quad (\text{B35})$$

with

$$C'_{i_1 \cdots i_r \cdots i_n} = C_{i_1 \cdots i_r \cdots i_n} - \sum_{i=2}^n F^{\gamma_1 \cdots \gamma_t}_{i_1 \cdots i_r i_p \cdots i_q \cdots i_n} D_{i_p}^{\gamma_2} \cdots D_{i_q}^{\gamma_t}, \quad (\text{B36})$$

and

$$\sum_{r=1}^n C'_{i_1 \cdots i_r \cdots i_n} D_{i_r}^{\gamma} = \sum_{r=1}^n C_{i_1 \cdots i_r \cdots i_n} D_{i_r}^{\gamma},$$

on account of Eq. (B34).

Q.E.D.

LEMMA B4. *Let a set of local functionals $B_{i_1 \cdots i_r \cdots i_n}^{\rho\alpha}[\Phi]$ ($1 \leq r \leq n$) and $A_{i_1 \cdots i_n k}^{\rho}[\Phi]$ satisfy*

$$\sum_{r=1}^n B_{i_1 \cdots i_r \cdots i_n}^{\rho\alpha} D_{i_r}^{\alpha} = A_{i_1 \cdots i_n k}^{\rho} \mathcal{L}_{0,k}. \quad (\text{B37})$$

Then, there exist local functionals $C_{i_1 \cdots i_r \cdots i_n k}^{\rho\alpha}[\Phi]$ such that

$$\sum_{r=1}^n B_{i_1 \cdots i_r \cdots i_n}^{\rho\alpha} D_{i_r}^{\alpha} = \sum_{r=1}^n C_{i_1 \cdots i_r \cdots i_n k}^{\rho\alpha} \mathcal{L}_{0,k} D_{i_r}^{\alpha}. \quad (\text{B38})$$

A, B, and C may carry additional indices.

Proof of B4. Since $\mathcal{L}_{0,k}$ and D_i^{α} have definite dimensions, we may assume that A and B have also. We assume that A begins with a term Φ^{m-1} and B with Φ^m . We expand Φ^{m-1} terms in A and Φ^m terms in B in powers of independent momenta associated with the points $(i_1 \cdots i_r \cdots i_n)$:

$$A_{i_1 \cdots i_n k}^{\rho(m-1)} = \bar{A}_{i_1 \cdots i_n k}^{\rho(m-1)} + \sum \partial_{i_p}^{\alpha} \bar{A}_{i_1 \cdots i_r i_p \cdots i_n}^{\rho\alpha(m-1)} + \sum_{p,q} \partial_{i_p}^{\alpha} \partial_{i_q}^{\beta} A_{i_1 \cdots i_r i_p i_q \cdots i_n}^{\rho\alpha\beta(m-1)} + \cdots. \quad (\text{B39})$$

$$B_{i_1 \cdots i_r \cdots i_n}^{\rho(m)} = \bar{B}_{i_1 \cdots i_r \cdots i_n}^{\rho(m)} + \sum_p \partial_{i_p}^{\beta} B_{i_1 \cdots i_r i_p \cdots i_n}^{\rho\beta(m)} + \sum_{p,q} \partial_{i_p}^{\beta} \partial_{i_q}^{\gamma} B_{i_1 \cdots i_r i_p i_q \cdots i_n}^{\rho\beta\gamma(m)} + \cdots. \quad (\text{B40})$$

We now compare the coefficients of Φ^m in Eq. (B37) and thus obtain,

$$\sum_{r=1}^n B_{i_1 \dots i_r \dots i_n, \{j\}}^{\rho\alpha} \partial_{i_r}^\alpha = \sum_{s=1}^m A_{i_1 \dots i_n k, j_1 \dots j_s}^\rho D_{kj_s}, \quad (\text{B41})$$

with

$$B_{i_1 \dots i_r \dots i_n, \{j\}}^{\rho\alpha} \equiv B_{i_1 \dots i_r \dots i_n, j_1 \dots j_m}^{\rho\alpha}. \quad (\text{B42})$$

We substitute the expansions of Eqs. (B39) and (B40) in Eq. (B41) and compare coefficients of various independent polynomials of momenta and obtain a set of equations:

$$\begin{aligned} \sum_s \bar{A}_{i_1 \dots i_n k, j_1 \dots j_s}^\rho D_{kj_s} &= 0, \\ \sum_s A_{i_1 \dots i_p \dots i_n k, j_1 \dots j_s}^{\rho\alpha} D_{kj_s} &= \bar{B}_{i_1 \dots i_p \dots i_n, \{j\}}^{\rho\alpha} \\ &\dots\dots\dots \\ \sum_s A_{i_1 \dots i_p \dots i_n k, j_1 \dots j_s}^{\rho\alpha_1 \dots \alpha_t} D_{kj_s} &= (t!) S[\alpha_1 \dots \alpha_t] B_{i_1 \dots i_p \dots i_n, \{j\}}^{\rho\alpha_1 \dots \alpha_t} \end{aligned} \quad (\text{B43})$$

etc.

We then define

$$\begin{aligned} (m-1)! C_{i_1 \dots i_r \dots i_n k}^{\rho\alpha(m-1)} &= \left[A_{i_1 \dots i_r \dots i_n k, j_1 \dots j_{m-1}}^{\rho\alpha(m-1)} + \sum_p \partial_{i_p}^\beta A_{i_1 \dots i_r i_p \dots i_n k, j_1 \dots j_{m-1}}^{\rho\alpha\beta(m-1)} \right. \\ &\quad \left. + \sum_{p,q} \partial_{i_p}^\beta \partial_{i_q}^\gamma A_{i_1 \dots i_r i_p i_q \dots i_n k, j_1 \dots j_{m-1}}^{\rho\alpha\beta\gamma} + \dots \right] \left(\prod_{s=1}^{m-1} \Phi_{j_s} \right). \end{aligned} \quad (\text{B44})$$

From Eqs. (B43) it follows that

$$\sum_{r=1}^n B_{i_1 \dots i_r \dots i_n, \{j\}}^{\rho\alpha(m)} \partial_{i_r}^\alpha = \sum_{r=1}^n C_{i_1 \dots i_r \dots i_n k}^{\rho\alpha(m-1)} \partial_{i_r}^\alpha D_{kj} \Phi_j. \quad (\text{B45})$$

We define,

$$A_{i_1 \dots i_n k}^{\rho[m]} = A_{i_1 \dots i_n k}^\rho - \sum_{r=1}^n C_{i_1 \dots i_r \dots i_n k}^{\rho\alpha(m-1)} D_{i_r}^\alpha \quad (\text{B46})$$

and

$$B_{i_1 \dots i_r \dots i_n}^{\rho\alpha[m+1]} = B_{i_1 \dots i_r \dots i_n}^{\rho\alpha} - C_{i_1 \dots i_r \dots i_n k}^{\rho\alpha(m-1)} \mathcal{L}_{0,k}. \quad (\text{B47})$$

It is clear that $A'^{\rho[m]}$ and $B'^{\rho\alpha[m+1]}$ may contain terms that go as Φ^{m-1} and Φ^m , respectively. But they satisfy

$$A'_{i_1 \dots i_n k}{}^{\rho[m]} D_{kj} \Phi_j = 0 + O[\Phi^{m+1}] \quad (\text{B48})$$

and

$$\sum_{r=1}^n B'_{i_1 \dots i_r k}{}^{\rho\alpha[m+1]} \partial_{i_r}^\alpha = 0 + O[\Phi^{m+2}] \quad (\text{B49})$$

From the proof of Lemmas A1 and A2, we know that given Eq. (B48) we can always construct a local functional $H_{i_1 \dots i_n k}^\rho[\Phi]$ such that

$$H_{i_1 \dots i_n k}^\rho \mathcal{L}_{0,k} = 0 \quad (\text{B50})$$

$$H_{i_1 \dots i_n k}^\rho = A'_{i_1 \dots i_n k}{}^{\rho[m]'} + O[\Phi^m]. \quad (\text{B51})$$

From the proof of Lemma (B1) we know that there exists a local functional $J_{i_1 \dots i_r k}^{\rho\alpha}[\Phi]$ such that

$$\sum_r J_{i_1 \dots i_r k}^{\rho\alpha} D_{i_r}^\alpha = 0 \quad (\text{B52})$$

and

$$J_{i_1 \dots i_r k}^{\rho\alpha} = B_{i_1 \dots i_r k}^{\rho\alpha} + O[\Phi^{m+1}]. \quad (\text{B53})$$

Then we form,

$$A^{\rho[m]} = A^{\rho[m]'} - H^\rho \sim O[\Phi^m] \quad (\text{B54})$$

$$B^{\rho\alpha[m+1]} = B^{\rho\alpha[m+1]'} - J^\rho \sim O[\Phi^{m+1}], \quad (\text{B55})$$

which satisfy Eq. (B37) viz.

$$A_{i_1 \dots i_n k}^{\rho[m]} \mathcal{L}_{0,k} = \sum_r B_{i_1 \dots i_r k}^{\rho\alpha[m+1]} D_{i_r}^\alpha. \quad (\text{B56})$$

This process can obviously be repeated. Each step increases the power of Φ in A and B by one; while their dimension remains constant. The process terminates when the power of Φ exceeds their dimension. The structure of Eq. (B47) and its analogs lead us to the statement of Eq. (B38).

We consider a local polynomial functional $Q_{iA}[\Phi, \omega]$. Here, A denotes the additional indices (if any) collectively. Let the power of ω in Q_{iA} be p . Let $D = \dim(Q_{iA})$. We define $\dim'(Q_{iA}) = D - p$.

LEMMA B5. *Let*

$$(\mathcal{G}Q_{iA}) \mathcal{L}_{0,i} = 0. \quad (\text{B57})$$

Then, there exists a local functional $P_{iA}[\Phi, \omega]$ such that

$$\mathcal{G}(Q_{iA} + P_{iA}) = 0, \quad (\text{B58})$$

$$P_{iA} \mathcal{L}_{0,i} = 0. \quad (\text{B59})$$

We note that Q_{iA} and P_{iA} have the same power of ω and the same dimensions.

In other words, given Eq. (B57), one can redefine Q_{iA} such that $\mathcal{G}Q_{iA} = 0$ without altering $Q_{iA} \mathcal{L}_{0,i}$.

We shall prove the lemma by induction. We shall omit the additional indices A whenever unessential.

Proof of B5. According to Lemma A2,

$$\mathcal{G}Q_{iA} = X_A{}^\gamma D_{i^\gamma} + Y_{ijA} \mathcal{L}_{0,j}, \quad (\text{B60})$$

with

$$Y_{ij} = \sum_{r=0}^n Y_{ij i_1 \dots i_r}^{(r)} \mathcal{L}_{0,i_1} \dots \mathcal{L}_{0,i_r}, \quad (\text{B61})$$

with

$$S[i_1 \dots i_{r+2}] Y_{i_1 \dots i_{r+2}}^{(r)} = 0. \quad (\text{B62})$$

(i) Suppose that $\dim'(Q_{iA}) < 3$. Then, $Y_{ij} = 0$ and the proposition is true for $\dim'(Q_{iA}) < 3$ by virtue of Lemma B2.

(ii) Let us assume that the proposition is true for $\dim'(Q_{iA}) < N$, for all possible A and all possible $p \geq 0$.

We wish now to prove it for $\dim'(Q_{iA}) = N$. From Eq. (B60)

$$(\mathcal{G}X_\gamma) D_{i^\gamma} = -\mathcal{G}(Y_{ij})' \mathcal{L}_{0,j}. \quad (\text{B63})$$

According to Lemma B4, there exists a local functional W_{rj} such that

$$\mathcal{G}X_\gamma = W_{\gamma j} \mathcal{L}_{0,j} \quad (\text{B64})$$

or

$$\mathcal{G}(W_{\gamma j}) \mathcal{L}_{0,j} = 0. \quad (\text{B65})$$

Since $\dim'(W_{\gamma j}) = \dim(X'_\gamma) - 3 = N - 4$, we can apply the proposition of this lemma to Eq. (B65) to conclude that we can always redefine (if necessary) $W_{\gamma j}$ without altering $W_{\gamma j} \mathcal{L}_{0,j}$ such that

$$\mathcal{G}(W_{\gamma j}) = 0. \quad (\text{B66})$$

We can apply the Dixon–Taylor lemma and write

$$W_{\gamma j} = \mathcal{G}X_{\gamma j}. \quad (\text{B67})$$

Therefore, from Eqs. (B64) and (B67),

$$\mathcal{G}(X_\gamma - X_{\gamma j} \mathcal{L}_{0,j}) = 0$$

or

$$X_\gamma = X_{\gamma j} \mathcal{L}_{0,j} + X_\gamma^0 \quad (\text{B68})$$

with

$$\mathcal{G}X_\gamma^0 = 0. \quad (\text{B69})$$

Substituting Eq. (B68) in Eq. (B60), we obtain,

$$\mathcal{G}Q_i = X_\gamma^0 D_i^\gamma + (X_{\gamma j} D_i^\gamma + Y_{ij}) \mathcal{L}_{0,j} \quad (\text{B70})$$

where from Eq. (B69)

$$\mathcal{G}(X_\gamma^0) D_i^\gamma = 0. \quad (\text{B71})$$

By Lemma B2, there exists a local functional T_γ^0 , such that

$$X_\gamma^0 = \mathcal{G}(T_\gamma^0) \quad (\text{B72})$$

or

$$\mathcal{G}(Q_i - T_\gamma^0 D_i^\gamma) = (X_{\gamma j} D_i^\gamma + Y_{ij}) \mathcal{L}_{0,j} \quad (\text{B73})$$

and thus

$$\mathcal{G}(X_{\gamma j} D_i^\gamma + Y_{ij}) \mathcal{L}_{0,j} = 0. \quad (\text{B74})$$

Further, $\dim'(X_{\gamma j} D_i^\gamma + Y_{ij}) = N - 3$. Therefore, we may apply the proposition to conclude that there exists a local functional P_{ij} such that

$$P_{ij} \mathcal{L}_{0,j} = 0 \quad (\text{B75})$$

and

$$\mathcal{G}(X_{\gamma j} D_i^\gamma + Y_{ij} + P_{ij}) = 0. \quad (\text{B76})$$

From Lemma A2, we may express P_{ij} as

$$P_{ij} = X_{\gamma i} D_j^\gamma + \sum_{r=1} P_{ij k_1 \dots k_r}^{(r)} \mathcal{L}_{0,k_1} \dots \mathcal{L}_{0,k_r} \quad (\text{B77})$$

with

$$S[i_1 \dots i_{r+2}] P_{i_1 \dots i_{r+2}}^{(r)} = 0. \quad (\text{B78})$$

We define

$$Y_{i_1 i_2 \dots i_{r+2}}^{(r)} = Y_{i_1 i_2 \dots i_{r+2}}^{(r)} + P_{i_1 i_2 \dots i_{r+2}}^{(r)}. \quad (\text{B79})$$

Then, on account of Eqs. (B78) and (B62)

$$S[i_1 \dots i_{r+2}] Y_{i_1 \dots i_{r+2}}^{(r)} = 0. \quad (\text{B80})$$

and

$$Y_{ij} \mathcal{L}_{0,j} = Y'_{ij} \mathcal{L}_{0,j} \equiv \sum_{r=1}^n Y_{ij i_1 \dots i_r}^{(r)} \mathcal{L}_{0,i_1} \cdots \mathcal{L}_{0,i_r} \mathcal{L}_{0,j}. \quad (\text{B81})$$

We may thus write Eq. (B76) as

$$\mathcal{G}(X_{vj} D_i^\gamma + X'_{vi} D_j^\gamma + Y'_{ij}) = 0. \quad (\text{B82})$$

We now define

$$\begin{aligned} C_{ij}^{(0)} &= \bar{S}[ij](X_{vj} D_i^\gamma + X'_{vi} D_j^\gamma + Y'_{ij}) \\ &= \tfrac{1}{2}[X_{vj} D_i^\gamma + X'_{vi} D_j^\gamma - X_{vj} D_i^\gamma - X'_{vi} D_j^\gamma] + Y_{ij}^{(0)} \\ &\quad + \sum_{r=1}^n \bar{S}[ij] Y_{ij i_1 \dots i_r}^{(r)} \mathcal{L}_{0,i_1} \cdots \mathcal{L}_{0,i_r}, \end{aligned} \quad (\text{B83})$$

and from Eq. (B82) it satisfies

$$\mathcal{G}(C_{ij}^{(0)}) = 0, \quad S[ij] C_{ij}^{(0)} = 0. \quad (\text{B84})$$

By the Dixon–Taylor lemma, there exists a local functional $U_{ij}^{(0)}$ such that

$$C_{ij}^{(0)} = \mathcal{G}U_{ij}^{(0)}, \quad S[ij] U_{ij}^{(0)} = 0. \quad (\text{B85})$$

Using Eqs. (B85) and (B83) in Eq. (B73), it may be written as

$$\begin{aligned} \mathcal{G}[Q_i - T_v^0 D_i^\gamma - U_{ij}^{(0)} \mathcal{L}_{0,j}] &= \{\tfrac{1}{2}[X_{vj} + X'_{vi}] D_i^\gamma + (i \leftrightarrow j)\} + S[ij] Y'_{ij} \mathcal{L}_{0,j} \\ &= \{X''_{vj} D_i^\gamma + X''_{vi} D_j^\gamma + S[ij] Y'_{ij}\} \mathcal{L}_{0,j} \end{aligned} \quad (\text{B86})$$

with

$$X''_{vj} = \tfrac{1}{2}(X_{vj} + X'_{vi}). \quad (\text{B87})$$

Further, from Eq. (B82)

$$\mathcal{G}\{X''_{vj} D_i^\gamma + X''_{vi} D_j^\gamma + S[ij] Y'_{ij}\} = S[ij] \mathcal{G}[X_{vj} D_i^\gamma + X'_{vi} D_j^\gamma + Y'_{ij}] = 0. \quad (\text{B88})$$

We note that $S[ij] Y'_{ij}$ contains at least one factor of $\mathcal{L}_{0,k}$, since $S[ij] Y_{ji}^{(0)} = 0$. Let

$$S[ij] Y'_{ij} = -Z_{ijk} \mathcal{L}_{0,k}. \quad (\text{B89})$$

Thus, Eq. (B88) can be written as

$$\mathcal{G}(X''_{vj}) D_i^\gamma + \mathcal{G}(X''_{vi}) D_j^\gamma = \mathcal{G}(Z_{ijk}) \mathcal{L}_{0,k}. \quad (\text{B90})$$

Applying Lemma B4 to Eq. (B90), we learn that there exist local functionals $C_{\gamma jk}$ such that

$$\mathcal{G}(X''_{\gamma j}) D_i^\gamma + \mathcal{G}(X''_{\gamma i}) D_j^\gamma = (C_{\gamma jk} D_i^\gamma + C_{\gamma ik} D_j^\gamma) \mathcal{L}_{0,k} = Z_{ijk} \mathcal{L}_{0,k}. \quad (\text{B91})$$

Thus,

$$\mathcal{G}(C_{\gamma jk} \mathcal{L}_{0,k}) D_i^\gamma + \mathcal{G}(C_{\gamma ik} \mathcal{L}_{0,k}) D_j^\gamma = 0. \quad (\text{B92})$$

We apply Lemma B3 to Eq. (B92) to conclude that we can choose $C_{\gamma jk} \mathcal{L}_{0,k}$ such that

$$\mathcal{G}(C_{\gamma jk}) \mathcal{L}_{0,k} = 0, \quad (\text{B93})$$

without altering $(C_{\gamma jk} D_i^\gamma + C_{\gamma ik} D_j^\gamma) \mathcal{L}_{0,k}$. Further, $\dim'(C_{\gamma jk}) = \dim'(X_{\gamma j}) = N - 3$. Thus, we may further apply the proposition to conclude that we may redefine $C_{\gamma jk}$ further so that

$$\mathcal{G}(C_{\gamma jk}) = 0 \quad (\text{B94})$$

with $C_{\gamma jk} \mathcal{L}_{0,k}$ unchanged. By the Dixon–Taylor lemma, there exists a local functional $W_{\gamma jk}$ such that

$$C_{\gamma jk} = \mathcal{G}(W_{\gamma jk}). \quad (\text{B95})$$

We may write Eq. (B91) as,

$$\mathcal{G}[W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma - Z_{ijk}] \mathcal{L}_{0,k} = 0. \quad (\text{B96})$$

Since $\dim'[W_{\gamma jk} D_i^\gamma + (i \leftrightarrow j) - Z_{ijk}] = \dim'(C_{\gamma jk}) = N - 3$, we may apply the proposition to conclude that there exists a local functional R_{ijk} such that

$$\mathcal{G}[W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma - Z_{ijk} - R_{ijk}] = 0, \quad (\text{B97})$$

with

$$R_{ijk} \mathcal{L}_{0,k} = 0. \quad (\text{B98})$$

From Lemma A2

$$R_{ijk} = -R'_{\gamma ij} D_k^\gamma + \sum_{r=1}^{r+1} R_{ijk i_1 \dots i_r}^{(r+1)} \mathcal{L}_{0,1} \dots \mathcal{L}_{0,i}, \quad (\text{B99})$$

with

$$S[i_3 \dots i_{r+3}] R_{i_1 \dots i_{r+3}}^{(r)} = 0. \quad (\text{B100})$$

We define

$$Z'_{ijk} = Z_{ijk} + \sum_{r=1}^{r+1} R_{ijk i_1 \dots i_r}^{(r+1)} \mathcal{L}_{0,i_1} \dots \mathcal{L}_{0,i_r}. \quad (\text{B101})$$

From the discussion under Eq. (B78), it is clear that we may define $Y_{i_1 \dots i_{r+2}}''^{(r)}$ ($r \geq 1$) so that

$$S[ij] Y_{ij}'' = -Z'_{ijk} \mathcal{L}_{0,k} = Z_{ijk} \mathcal{L}_{0,k}, \quad (\text{B102})$$

while $Y_{i_1 \dots i_{r+2}}''$ still satisfy

$$S[i_1 \dots i_{r+2}] Y_{i_1 \dots i_{r+2}}''^{(r)} = 0. \quad (\text{B103})$$

We thus have

$$\mathcal{G}[W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma + R'_{\gamma ij} D_k^\gamma - Z'_{ijk}] = 0. \quad (\text{B104})$$

We further note that Eq. (B91) implies, with the help of Eq. (B95)

$$\mathcal{G}[(X''_{\gamma j} - W_{\gamma jk} \mathcal{L}_{0,k}) D_i^\gamma + (i \leftrightarrow j)] = 0,$$

i.e.,

$$\mathcal{G}[(X''_{\gamma j} - W_{\gamma jk} \mathcal{L}_{0,k}) \mathcal{L}_{0,j}] D_i^\gamma = 0. \quad (\text{B105})$$

We may, now, express Eq. (B86) as

$$\begin{aligned} & \mathcal{G}[Q_i - T_\gamma^0 D_i^\gamma - U_{ij}^{(0)} \mathcal{L}_{0,j}] \\ &= (X''_{\gamma j} - W_{\gamma jk} \mathcal{L}_{0,k}) \mathcal{L}_{0,j} D_i^\gamma + [W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma + R'_{\gamma ij} D_k^\gamma - Z'_{ijk}] \mathcal{L}_{0,j} \mathcal{L}_{0,k}. \end{aligned} \quad (\text{B106})$$

On account of Eq. (B105) we may write

$$(X''_{\gamma j} - W_{\gamma jk} \mathcal{L}_{0,k}) \mathcal{L}_{0,j} D_i^\gamma = \mathcal{G}(T_\gamma^{(1)}) D_i^\gamma. \quad (\text{B107})$$

Further, Eq. (B104) yields

$$\mathcal{G}[\bar{S}[ijk]\{W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma + R'_{\gamma ij} D_k^\gamma - Z'_{ijk}\}] = 0. \quad (\text{B108})$$

Thus, there exists a local functional $U_{ijk}^{(1)}$ such that

$$S[ijk]\{W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma + R'_{\gamma ij} D_k^\gamma - Z'_{ijk}\} = \mathcal{G}(U_{ijk}^{(1)}) \quad (\text{B109})$$

with

$$\bar{S}[ij] U_{ijk}^{(1)} = 0, \quad S[ijk] U^{(1)}(ijk) = 0. \quad (\text{B110})$$

Using Eq. (B107) and (B109) in Eq. (B106) we get

$$\begin{aligned} & \mathcal{G}[Q_i - (T_\gamma^0 + T_\gamma^1) D_i^\gamma - U_{ij}^0 \mathcal{L}_{0,j} - U_{ijk}^{(1)} \mathcal{L}_{0,j} \mathcal{L}_{0,k}] \\ &= \{S[ijk] X''_{\gamma jk} D_i^\gamma - S[ijk] Z'_{ijk}\} \mathcal{L}_{0,k} \mathcal{L}_{0,j} \\ &= \{S[ijk] X''_{\gamma jk} D_i^\gamma + S[ijk] \sum_{r=1}^n Y_{ijk i_1 \dots i_r}^{(r+1)} \mathcal{L}_{0,i_1} \dots \mathcal{L}_{0,i_r}\} \mathcal{L}_{0,j} \mathcal{L}_{0,k} \end{aligned} \quad (\text{B111})$$

where

$$S[ijk] X''_{\gamma jk} D_i^\gamma \equiv S[ijk] \{W_{\gamma jk} D_i^\gamma + W_{\gamma ik} D_j^\gamma + R'_{\gamma ij} D_k^\gamma\} \quad (\text{B112})$$

and

$$\mathcal{G}\{S[ijk] X''_{\gamma jk} D_i^\gamma + S[ijk] \sum_{r=1}^n Y''_{ijk i_1 \dots i_r} \mathcal{L}_{0, i_1} \dots \mathcal{L}_{0, i_r}\} = 0 \quad (\text{B113})$$

This process clearly can be continued to the end with the help of Lemmas B3 and B4. In the end, one would obtain

$$\mathcal{G} \left[\mathcal{Q}_i - \sum_{r=0}^{n+1} T_\gamma^{(r)} D_i^\gamma - \sum_{r=0}^n U_{ii_2 \dots i_{r+2}}^{(r)} \mathcal{L}_{0, i_2} \dots \mathcal{L}_{0, i_{r+2}} \right] = 0 \quad (\text{B114})$$

$$S[i_1 \dots i_{r+2}] U_{i_1 \dots i_{r+2}}^{(r)} = 0. \quad (\text{B115})$$

Further,

$$\left[\sum_{r=0}^{n+1} T_\gamma^{(r)} D_i^\gamma - \sum_{r=0}^n U_{ii_2 \dots i_{r+2}}^{(r)} \mathcal{L}_{0, i_2} \dots \mathcal{L}_{0, i_{r+2}} \right] \mathcal{L}_{0, i} = 0. \quad (\text{B116})$$

Q.E.D.

APPENDIX C

In Appendix B we have applied the Dixon–Taylor lemma to the operator $\mathcal{G}$ of Eq. (4.28). One may apply the lemma since the Eq. (4.11) holds for $A = \omega_\eta(\delta/\delta\Phi_\eta^L)$, when X is proportional to at least one ghost field ω . Here, we have decomposed $\mathcal{G}$ as

$$\mathcal{G} = A + gB, \quad \mathcal{G}^2 = 0. \quad (\text{C1})$$

However, because of the presence of the derivative with respect to Φ_η^L in A , one does not know, as mentioned in Section IV, whether X of Eq. (4.11) can be chosen to be local, given that Y is local. It should be emphasized that there are instances in which the Dixon–Taylor construction can be used to construct J of Eq. (4.10), however, it is impossible to choose J to be local even though H of Eq. (4.9) is local. We shall show that *in the cases of our interest* in Appendix B, it is possible to choose the solution “ J ” to be local by showing that in the cases of our interest the Eq.

$$BY = 0, \quad Y \text{ local} \quad (\text{C2})$$

implies that Y may be expressed as

$$Y = BX + AW, \quad (\text{C3})$$

for some local X and W . (In particular, W may be zero, which will be the case in all applications of Appendix B so that the Dixon-Taylor construction can be carried out "backwards.") This way, the locality of the solution is clear. We consider the Eq. (C2), where Y carried a free index, say 1. The following relations are easily verified.

$$B(\Phi_i) = 0, \quad B(t_{ij}^\alpha) = 0, \quad B(f^{\alpha\beta\gamma}) = 0. \quad (C4)$$

The last two imply

$$B(g_{\mu\nu}) = 0. \quad (C5)$$

Further,

$$B(\delta_{ij}) = 0, \quad B(\delta_{\alpha\beta}) = 0. \quad (C6)$$

Also, we define

$$B(\delta^4(x - y)) = 0. \quad (C7)$$

However,

$$B(\partial_i^\alpha) = g(\omega\partial)_j t_{ij}^\alpha \neq 0, \quad (\omega\partial)_j \equiv \omega_\alpha \partial_j^\alpha, \quad (C8)$$

and

$$B(\omega_\alpha) = \frac{1}{2} g_{\alpha\beta\gamma} \omega_\beta \omega_\gamma \neq 0, \quad (C9)$$

nevertheless,

$$B(\omega\partial)_i = 0. \quad (C10)$$

Without loss of generality, we may consider $Y(\omega, \Phi)$, which involves a fixed number of ω and Φ

$$Y_i[\omega, \Phi] = \omega_{\alpha_1} \cdots \omega_{\alpha_n} Y_{l, k_1 \cdots k_m}^{\alpha_1 \cdots \alpha_n} \Phi_{k_1} \cdots \Phi_{k_m}. \quad (C11)$$

Then, Eq. (C2) together with $B(\Phi_i) = 0$ implies

$$B[\omega_{\alpha_1} \cdots \omega_{\alpha_n} Y_{l, k_1 \cdots k_m}^{\alpha_1 \cdots \alpha_n}] = 0. \quad (C12)$$

Thus, it is sufficient to consider Y not containing fields Φ but carrying an arbitrary number of indices. Now, the operator B does not alter the number of derivatives in a functional. Hence, we may assume that Y contains a fixed number of derivatives.

We shall first consider the case in which Y does not contain any derivatives. The considerations involved in this case will be useful in the case in which Y contains derivatives. In this case, $Y_{l, k_1 \cdots k_m}^{\alpha_1 \cdots \alpha_n}$ consists only of the B -invariants, so that Eq. (C12) implies

$$B[\omega_{\alpha_1} \cdots \omega_{\alpha_n}] Y^{\alpha_1 \cdots \alpha_n} = 0, \quad (C13)$$

where we have suppressed the extra free indices. Thus,

$$f_{\eta\alpha_n\alpha_{n+1}}\omega_{\alpha_1}\cdots\omega_{\alpha_{n+1}}Y^{\alpha_1\cdots\alpha_{n+1}\eta} = 0. \quad (C14)$$

Thus, the totally antisymmetric part of $f_{\eta\alpha_n\alpha_{n+1}}Y^{\alpha_1\cdots\alpha_{n+1}\eta}$ vanishes. It consists of a sum of $((n+1)/2) = n(n+1)/2$ terms, since the above is already antisymmetric in $(\alpha_1 \cdots \alpha_{n-1})$ and in (α_n, α_{n+1}) . We write one of these terms as

$$\begin{aligned} f_{\eta\alpha_n\alpha_{n+1}}Y^{\alpha_1\cdots\alpha_{n+1}\eta} &= -f_{\eta\alpha_{n-1}\alpha_n}Y^{\alpha_{n+1}\alpha_1\cdots\alpha_{n-1}\eta} + \cdots \\ &\quad -f_{\eta\alpha_1\alpha_2}Y^{\alpha_3\cdots\alpha_n\alpha_{n+1}\eta} + \cdots, \end{aligned} \quad (C15)$$

where we have shown two typical terms on the right-hand side. Since Y does not contain derivatives, each term contains a product of same δ functions that can be extracted. Thus, in Eq. (10), we shall assume that the terms have no space-time dependence. We multiply Eq. (C15) by $f_{\xi\alpha_n\alpha_{n+1}}$ and use,

$$f_{\eta\alpha_n\alpha_{n+1}}f_{\xi\alpha_n\alpha_{n+1}} = \delta_{\eta\xi}C_2, \quad (C16)$$

so that,

$$\begin{aligned} C_2Y^{\alpha_1\cdots\alpha_{n-1}\xi} &= -f_{\xi\sigma\lambda}f_{\eta\alpha_{n-1}\sigma}Y^{\lambda\alpha_1\cdots\alpha_{n-2}\eta} + \cdots \\ &\quad -f_{\eta\alpha_1\alpha_2}f_{\xi\sigma\lambda}Y^{\alpha_3\cdots\alpha_{n-1}\sigma\lambda\eta} + \cdots, \end{aligned} \quad (C17)$$

and thus,

$$\begin{aligned} C_2\omega_{\alpha_1}\cdots\omega_{\alpha_{n-1}}\omega_{\xi}Y^{\alpha_1\cdots\alpha_{n-1}\xi} &= -(f_{\xi\sigma\lambda}f_{\eta\alpha_{n-1}\sigma}\omega_{\alpha_{n-1}}\omega_{\xi})Y^{\lambda\alpha_1\cdots\alpha_{n-2}\eta} + \cdots \\ &\quad -f_{\eta\alpha_1\alpha_2}\omega_{\alpha_1}\omega_{\alpha_2}f_{\xi\sigma\lambda}Y^{\alpha_3\cdots\alpha_{n-1}\sigma\lambda\eta} + \cdots. \end{aligned} \quad (C18)$$

In the first term on the right-hand side of Eq. (C18), we use,

$$\begin{aligned} \omega_{\alpha_{n-1}}\omega_{\xi}f_{\xi\sigma\lambda}f_{\eta\alpha_{n-1}\sigma} &= \frac{1}{2}\omega_{\alpha_{n-1}}\omega_{\xi}\{f_{\xi\sigma\lambda}f_{\eta\alpha_{n-1}\sigma} - (\xi \leftrightarrow \alpha_{n-1})\} \\ &= \frac{1}{2}\omega_{\alpha_{n-1}}\omega_{\xi}f_{\alpha_{n-1}\xi\kappa}f_{\lambda\kappa\eta}, \end{aligned} \quad (C19)$$

and thus, we can write Eq. (C18) as

$$\omega_{\alpha_1}\cdots\omega_{\alpha_n}Y^{\alpha_1\cdots\alpha_{n-1}} = \frac{1}{2}f_{\gamma\delta\kappa}\omega_r\omega_{\delta}\{A_1f_{\kappa\lambda\eta}Y^{\lambda\alpha_1\cdots\alpha_{n-2}\eta} + A_2f_{\xi\sigma\lambda}Y^{\alpha_1\cdots\alpha_{n-3}\sigma\lambda\eta}\}, \quad (C20)$$

where A_1 and A_2 are some numbers depending on n and C_2 .

Even though the argument following Eq. (C14) does not go through in the case when n equals the number of generators of the group, the relation of the form of Eq. (C20) still holds (with $A_2 = 0$). This can be seen by explicitly working out the right-hand side of Eq. (C20) for this case. As we shall see later, we shall

need the relation only for the case of $SU(2)$ (i.e., three ghosts) in which case we see that

$$\begin{aligned} \frac{1}{2}\epsilon_{\gamma\delta\kappa}\omega_\gamma\omega_\delta\epsilon_{\lambda\kappa\eta}Y^{\lambda\alpha\eta}\omega_\alpha &= \frac{1}{2}\omega_\gamma\omega_\delta\omega_\alpha\{-\delta_{\gamma\lambda}\delta_{\delta\eta} + \delta_{\gamma\eta}\delta_{\delta\lambda}\}Y^{\lambda\alpha\eta} \\ &= Y^{\alpha\beta\gamma}\omega_\alpha\omega_\beta\omega_\gamma. \end{aligned} \quad (C21)$$

Using Eq. (C20) we define an $X^{\kappa\alpha_1\cdots\alpha_{n-2}}$ by,

$$Y^{\alpha_1\cdots\alpha_n}\omega_{\alpha_1}\cdots\omega_{\alpha_n} \equiv \frac{1}{2}f_{\gamma\delta\kappa}\omega_\gamma\omega_\delta X^{\kappa\alpha_1\cdots\alpha_{n-2}}\omega_{\alpha_1}\cdots\omega_{\alpha_{n-2}}. \quad (C22)$$

$X^{\kappa\alpha_1\cdots\alpha_{n-2}}$ is clearly antisymmetric in its last $(n-2)$ indices. Therefore, it can be expressed as a sum of two terms each having either one of the permutation symmetries of Young tableaux in Fig. 2. We write

$$X = X' + X'', \quad (C23)$$

where X' is totally antisymmetric in $(\kappa\alpha_1\cdots\alpha_{n-2})$. Substituting the decomposition of Eq. (C23) in Eq. (C22), we get a corresponding decomposition for Y

$$Y = Y' + Y''. \quad (C24)$$

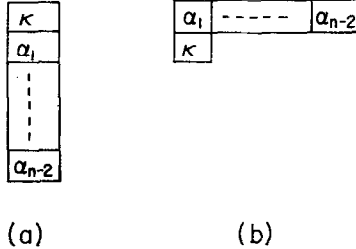


FIG. 2. Young Tableaux for the permutation symmetries of X' and X'' of Eq. (C.23).

It is clear that Y' may be expressed as,

$$Y'^{\alpha_1\cdots\alpha_n}\omega_{\alpha_1}\cdots\omega_{\alpha_n} \simeq B(\omega_\kappa X^{\kappa\alpha_1\cdots\alpha_{n-2}}\omega_{\alpha_1}\cdots\omega_{\alpha_{n-2}}), \quad (C25)$$

where $\simeq$ means equal within a numerical factor, and

$$BY'' = 0. \quad (C26)$$

Thus, showing that $BY = 0 \Rightarrow Y = BX$ (X local), in our cases of interest amounts to examining B -invariants of the form Y'' (related to X'' with the permutation symmetry of the Young tableau of Fig. (2b) by Eq. (C22)), and showing that such B -invariants do not occur in the cases of our interest. We define such B -invariants as anomalous invariants. The anomalous invariants cannot be expressed as BZ

since the X -function of Eq. (C22) for BZ for any Z always can be chosen to be antisymmetric (and in fact proportional to Z). Since the classification of B -invariants depends entirely on the indices of Y contracted with ghosts; it follows that:

(i) If $Y_i[\omega, \Phi]$ is an anomalous invariant, so is $Y_i[\Phi] T_i[\Phi]$, where $T_i[\Phi]$ is a local functional not containing derivatives.

(ii) If $Y^{(1)}$ and $Y^{(2)}$ are anomalous invariants, so is $Y^{(1)} + Y^{(2)}$.

(iii) If $Y_i[\omega, \Phi]$ is an anomalous invariant, and $Y_i[\omega, \Phi] = K_{ij}[\omega, \Phi] L_j[\Phi]$, where L is local, $K_{ij}[\omega, \Phi]$ is also an anomalous invariant, (i.e., the part of K that contributes to $K_{ij}L_j$).

Therefore, let us consider Eq. (C22), where Y is an anomalous invariant. Using $B(Y) = 0$, we get,

$$Z_{\kappa\alpha_1 \dots \alpha_{n-2}} X^{\kappa\alpha_1 \dots \alpha_{n-2}} = 0, \quad (\text{C27})$$

with,

$$Z_{\kappa\alpha_1 \dots \alpha_{n-2}} = A[\alpha_1 \dots \alpha_{n-2}] \{ f_{\kappa\gamma\delta} \omega_\gamma \omega_\delta f_{\alpha_1 \varepsilon \eta} \omega_\varepsilon \omega_\eta \omega_{\alpha_2} \dots \omega_{\alpha_{n-2}} \}, \quad (\text{C28})$$

and thus, Z as well as X has the permutation symmetry of the Young tableau of Fig. (2b). Further, for $n = 3$, Z satisfies

$$\delta^{\alpha\kappa} Z_{\kappa\alpha} = 0, \quad (\text{C29})$$

on account of the Jacobi identity.

An examination of Lemmas B3 and B5, shows that we have applied the Dixon-Taylor lemma for $\mathcal{G}$ in the cases when H of Eq. (4.8) contains two, three, or four ghosts. Therefore, we shall need to examine only these possibilities.

(i) Two ghosts: Here, Eq. (C22) gives

$$\omega_\alpha \omega_\beta Y^{\alpha\beta} = \frac{1}{2} f_{\kappa\gamma\delta} \omega_\gamma \omega_\delta X^\kappa = B(\omega_\kappa X^\kappa),$$

without exceptions. Thus, there are no anomalous invariants containing two ghosts.

(ii) Three ghosts: Eqs. (C27) and (C29) imply that the traceless symmetric part of $X^{\kappa\alpha}$ (which is already given to be symmetric in (κ, α)) vanishes. Thus, $X^{\kappa\alpha}$ must be proportional to $\delta^{\kappa\alpha}$. This argument does not apply to the case of the group $SU(2)$ because there, Z vanishes identically. But in this case, it is clear that only the part of $X^{\kappa\alpha}$ proportional to $\delta^{\kappa\alpha}$ contributes to $f_{\kappa\gamma\delta} \omega_\gamma \omega_\delta X^{\kappa\alpha} \omega_\alpha$. Thus, such anomalous B -invariants have the disconnected form

$$f_{\alpha\beta\gamma} \omega_\alpha \omega_\beta \omega_\gamma X[\Phi],$$

where X carries all other indices that Y does.

(iii) Four ghosts: Here, we shall not need to know the form of the anomalous invariants. We have used the Dixon–Taylor lemma after Eq. (B32), where E' contains four ghosts. We shall show that E' can be chosen not to contain anomalous invariants. We note that in Eq. (B23) the terms containing no derivatives on the right-hand side arise entirely from the action of B on the terms in C not containing derivatives, since C is local. Thus, this term on the right-hand side of Eq. (B32) is expressible as BZ , and by definition, does not contain anomalous invariants. Therefore, it is clear that terms in E not containing derivatives may be chosen not to contain anomalous invariants. Since E' are a linear combination of product of E and D_i'' 's, it is clear that terms with no derivatives in E' do not contain anomalous invariants, if such terms in E do not.

Therefore, we need only to worry about the application of the lemma to terms containing three ghosts. Application to Eq. (B66) is a typical example that was repeated at each subsequent stage in Appendix B.

We note that in Eq. (B64), X_γ is a local functional containing two ghosts. Thus, terms in $W_{\gamma j} \mathcal{L}_{0,j}$ not containing derivatives do not contain the anomalous invariants as argued in the discussion in the case of four ghosts. Therefore, terms in $W_{\gamma j}$ not containing derivatives may be assumed not to contain anomalous invariants. The question then, is whether one needs to add such terms to $W_{\gamma j}$ so as to make Eq. (B66) valid. This will be so if

$$\begin{aligned} \mathcal{G}(W_{rj}[\Phi]) &= -\mathcal{G}(f_{\alpha\beta\gamma}\omega_\alpha\omega_\beta\omega_\gamma X[\Phi]) \\ &= -f_{\alpha\beta\gamma}\omega_\alpha\omega_\beta\omega_\gamma \mathcal{G}(X[\Phi]). \end{aligned} \quad (C30)$$

On the right-hand side, $X[\Phi]$ is a local functional with no derivatives in it. Thus,

$$\mathcal{G}(X[\Phi]) = AX[\Phi] = \omega_\eta \partial_i{}^\eta X_{ji}[\Phi].$$

Then, Eq. (C30) becomes

$$\mathcal{G}(W_{rj}[\Phi]) = -f_{\alpha\beta\gamma}\omega_\alpha\omega_\beta\omega_\gamma\omega_\eta \partial_i{}^\eta X_{ji}[\Phi]. \quad (C30a)$$

The terms in the left-hand side of Eq. (C30a) that contain one derivative can come from either the action of A on terms containing no derivatives ($\equiv \omega_\alpha\omega_\beta\omega_\gamma U^{\alpha\beta\gamma}[\Phi]$), or the action on B on terms containing one derivative ($\equiv \omega_\alpha\omega_\beta\omega_\gamma V_k^{\alpha\beta\gamma\delta} \partial_k{}^\delta$). Thus, we have to see if the following equation is possible:

$$\begin{aligned} & -\omega_\alpha\omega_\beta\omega_\gamma\omega_\eta \partial_i{}^\eta U_i^{\alpha\beta\gamma}[\Phi] + B(\omega_\alpha\omega_\beta\omega_\gamma) V_k^{\alpha\beta\gamma\delta}[\Phi] \partial_k{}^\delta + \omega_\alpha\omega_\beta\omega_\gamma V_k^{\alpha\beta\gamma\delta} B(\partial_k{}^\delta) \\ & = -f_{\alpha\beta\gamma}\omega_\alpha\omega_\beta\omega_\gamma\omega_\eta \partial_i{}^\eta X_{ji}[\Phi]. \end{aligned} \quad (C31)$$

From Eq. (C31), it is clear that $\partial_k{}^\delta$ must be contracted with a ghost. Then, letting

$$\omega_\alpha\omega_\beta\omega_\gamma V_k^{\alpha\beta\gamma\delta} \partial_k{}^\delta = \omega_\alpha \partial_i{}^\alpha \omega_\beta\omega_\gamma V_i^{\beta\gamma}[\Phi],$$

we get

$$-\omega_\alpha \omega_\beta \omega_\gamma \omega_\eta \partial_i^\eta U_{,i}^{\alpha\beta\gamma}[\Phi] + \omega_\alpha \partial_i^\alpha B(\omega_\beta \omega_\gamma) V_i^{\beta\gamma}[\Phi] = -f_{\alpha\beta\gamma} \omega_\alpha \omega_\beta \omega_\gamma \omega_\eta \partial_i^\eta X_{,i}.$$

But this is impossible, since $U^{\alpha\beta\gamma}$ does not contain a factor of the form $f_{\alpha\beta\gamma}$.

Next, we consider the case when Y contains derivatives. Let Y contain exactly r derivatives at least p ($0 \leq p \leq r$) of which are contracted with ω in the form $\omega_\alpha \partial_i^\alpha$. Therefore, BY consists of a sum of terms containing $(n+1)$ ghosts, r derivatives, at least p of which are contracted with ω 's. We write

$$\begin{aligned} Y_{l,}^{\alpha_1 \dots \alpha_n} \omega_{\alpha_1} \dots \omega_{\alpha_n} \\ = \omega_{\alpha_1} \dots \omega_{\alpha_{n-p}} (\omega \cdot \partial)_{j_1} \dots (\omega \cdot \partial)_{j_p} \partial_{k_1}^{\beta_1} \dots \partial_{k_{r-p}}^{\beta_{r-p}} Y_{l, j_1 \dots j_p, k_1 \dots k_{r-p}}^{\alpha_1 \dots \alpha_{n-p}, \beta_1 \dots \beta_{r-p}} + \dots \end{aligned} \quad (C32)$$

The terms in BY containing precisely p derivatives contracted with ghosts come from the action of B on $(\omega_{\alpha_1} \dots \omega_{\alpha_{n-p}})$ only. These terms must separately vanish. Thus,

$$0 = B(\omega_{\alpha_1} \dots \omega_{\alpha_{n-p}}) (\omega \cdot \partial)_{j_1} \dots (\omega \cdot \partial)_{j_p} \partial_{k_1}^{\beta_1} \dots \partial_{k_{r-p}}^{\beta_{r-p}} Y_{l, j_1 \dots j_p, k_1 \dots k_{r-p}}^{\alpha_1 \dots \alpha_{n-p}, \beta_1 \dots \beta_{r-p}}. \quad (C33)$$

We note that the momenta carried by all free space-time points except l are independent. Further, $Y'_{l, j_1 \dots j_p}$ is totally antisymmetrized in $(j_1 \dots j_p)$. Using these and comparing coefficients of independent moment a , we find,

$$B(\omega_{\alpha_1} \dots \omega_{\alpha_{n-p}}) Y_{l, j_1 \dots j_p, k_1 \dots k_{r-p}}^{\alpha_1 \dots \alpha_{n-p}, \beta_1 \dots \beta_{r-p}} = D, \quad (C34)$$

which is precisely the equation we solved earlier; with the result that (we suppress some indices here)

$$(\omega_{\alpha_1} \dots \omega_{\alpha_{n-p}} Y_{l, \dots}^{\alpha_1 \dots \alpha_{n-p}}) = B(\omega_{\alpha_1} \dots \omega_{\alpha_{n-p-1}} X_{l, \dots}^{\alpha_1 \dots \alpha_{n-p-1}}).$$

We now construct

$$Y_l^{(1)} = Y_l - B\omega_{\alpha_1} \dots \omega_{\alpha_{n-p-1}} X_{l, j_1 \dots j_p, k_1 \dots k_{r-p}}^{\alpha_1 \dots \alpha_{n-p-1}, \beta_1 \dots \beta_{r-p}} \partial_{k_1}^{\beta_1} \dots \partial_{k_{r-p}}^{\beta_{r-p}} (\omega \cdot \partial)_{j_1} \dots (\omega \cdot \partial)_{j_p}. \quad (C35)$$

Now, $Y_l^{(1)}$ satisfies $BY_l^{(1)} = 0$, and contains at least $(p+1)$ derivatives contracted with ghosts. This process can be obviously continued until the end [15].

Finally, we prove the locality of S_i of Eq. (4.22), which together with proof given by Dixon and Taylor in [5] completes the proof of the statement of Eq. (4.43). Here, we use the Dixon–Taylor lemma for functionals with one ghost. We thus have to show that

$$B(\omega_\alpha Y_i^\alpha[\Phi]) = 0 \quad (C36)$$

implies that for some local $X_i^\alpha[\Phi]$ and $Z_i^\alpha[\Phi]$,

$$\omega_\alpha Y_i^\alpha[\Phi] = B(X_i^\alpha[\Phi]) + A(Z_i^\alpha[\Phi]). \quad (C37)$$

From the discussion given earlier in this Appendix, we need to consider Y_i , which is independent of Φ , carrying an arbitrary number of additional indices in

$$B[\omega_\alpha Y_i^\alpha] = 0. \quad (C38)$$

Now, Y_i^α must contain at least one derivative for otherwise

$$B[\omega_\alpha Y_i^\alpha] = \frac{1}{2} g f_{\alpha\beta\gamma} \omega_\beta \omega_\gamma Y_i^\alpha = 0 \quad (C39)$$

implies $Y_i^\alpha \equiv 0$. Further, one of the derivatives in Y_i^α must be contracted with ω_α in the form $(\omega_\alpha \partial_m^\alpha)$, for, if we write

$$\omega_\alpha Y_i^\alpha = \omega_\alpha \partial_m^\alpha Y'_{mi}[\partial] + \omega_\alpha \partial_{k_1}^{\beta_1} \dots \partial_{k_n}^{\beta_n} Y''^{\alpha, \beta_1 \dots \beta_n}_{i, k_1 \dots k_n}, \quad (C40)$$

where the last term does not contain a factor $\omega_\alpha \partial_m^\alpha$, then, comparing coefficients of terms in Eq. (C38) not containing such a factor, we get

$$\frac{1}{2} g f_{\alpha\beta\gamma} \omega_\beta \omega_\gamma [\partial_{k_1}^{\beta_1} \dots \partial_{k_n}^{\beta_n} Y''^{\alpha, \beta_1 \dots \beta_n}_{i, k_1 \dots k_n}] = 0, \quad (C41)$$

showing that the last term in Eq. (C40) vanishes. Let us first consider the case in which Y'_{mi} does not contain derivatives. Then, remembering that Y'_{mi} is a local functional either [15]

$$Y'_{mi, \dots} = t_{mp}^\gamma Y''_{pi, \dots}, \quad (C42)$$

or

$$\omega_\alpha Y_i^\alpha[\Phi] \equiv \omega_\alpha \partial_m^\alpha Y'_{mi}[\Phi] = \omega_\alpha \partial_m^\alpha \{\Phi_m (\Phi_j \Phi_j)^p F_i[\Phi] \delta^4(\dots)\}, \quad (C43)$$

where, in $F[\Phi]$, all Φ are contracted with f 's or t 's. In the latter case, we may write

$$\begin{aligned} \omega_\alpha Y_i^\alpha[\Phi] &= -\omega_\alpha \partial_m^\alpha (\Phi_j \Phi_j)^{p+1} (\delta F_i[\Phi] / \delta \Phi_m) \delta^4(\dots) \\ &\quad + A\{(1/2(p+1))(\Phi_j \Phi_j)^{p+1} F_i[\Phi] \delta^4(\dots)\}, \end{aligned} \quad (C44)$$

where, the first term on the right-hand side is itself a B -invariant and has the form of the term in Eq. (C42). Thus, we need consider Y'_{mi} of the form of the right-hand side of Eq. (C42) only. Then,

$$\begin{aligned} \omega_\alpha \partial_m^\alpha Y'_{mi} &= \omega_\alpha \partial_m^\alpha t_{mp}^\gamma Y''_{pi} \\ &= (1/g) B(\partial_p^\gamma Y''_{pi}). \end{aligned} \quad (C45)$$

Equations (C44) and (C45) prove the result for this case. Finally, in the case when Y_{mi} contains derivatives, we write

$$\omega_\alpha Y_i^\alpha = \omega_\alpha \partial_m^\alpha \partial_{j_1}^{\beta_1} \dots \partial_{j_n}^{\beta_n} Y_{mi, j_1 \dots j_n}^{\beta_1 \dots \beta_n}, \quad (\text{C46})$$

where $Y_{j_1 \dots j_n}^{\beta_1 \dots \beta_n}$ has the appropriate symmetry. Then, $B(\omega_\alpha Y_i^\alpha) = 0$ implies

$$\omega_\alpha \partial_m^\alpha \omega_\gamma \partial_k^\gamma t_{kj_1}^{\beta_1} \partial_{j_2}^{\beta_2} \dots \partial_{j_n}^{\beta_n} Y_{mi, j_1 \dots j_n}^{\beta_1 \dots \beta_n} = 0, \quad (\text{C47})$$

i.e.,

$$t_{kj_1}^{\beta_1} Y_{mi, j_1 \dots j_n}^{\beta_1 \dots \beta_n} - (k \leftrightarrow m) = 0. \quad (\text{C48})$$

It is easy to verify that such a relation necessarily requires that the group index in m is attached to a t_{ml}^η . Then, Eq. (C46) can be written as

$$\omega_\alpha Y_i^\alpha = \omega_\alpha \partial_m^\alpha t_{ml}^\eta \partial_{j_1}^{\beta_1} \dots \partial_{j_n}^{\beta_n} K_{i, j_1 \dots j_n}^{\beta_1 \dots \beta_n \eta} \quad (\text{C49})$$

Then, Eq. (C48) implies that $K_{i, j_1 \dots j_n}^{\beta_1 \dots \beta_n \eta}$ is symmetric under the interchange of $(\beta_r, j_r) \leftrightarrow (\eta, l)$. Thus, we may write

$$\omega_\alpha Y_i^\alpha = B(\partial_l^\eta \partial_{j_1}^{\beta_1} \dots \partial_{j_n}^{\beta_n} K_{i, j_1 \dots j_n}^{\beta_1 \dots \beta_n \eta}) \quad (\text{C50})$$

proving the result.

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12. Here "solution" does not refer to the solution of Eq. (1.2), but to the solution obtained by the Dixon-Taylor construction.
13. See, for example, M. HAMERMESH, "Group Theory," p. 244. Addison-Wesley, Reading, Mass. 1962.
14. There may be gauge invariant operators which may be expressible in this form ($\mathcal{G}'F$). They are expressible as $(\delta\mathcal{L}_0/\delta A_i) F_i[A]$, where F_i is a covariant local functional, i.e., $\mathcal{G}F_i = 0$. We thank Dr. Zuber for pointing out this to us. In the case of twist two operators, however, there are no such gauge invariant operators.
15. Since we are interested in solving $\mathcal{G}W = 0$, i.e., $(A + gB)W = 0$, only those B -invariants that can occur in a $\mathcal{G}$ invariant W are of interest to us.